

Unit 7 – Hypothesis Testing

Introduction to Statistical Inference

One-Sample, Paired, and Two-Sample t -Tests

Scope of these notes

We begin with hypotheses, claims, significance, and decisions. We then study one-sample, paired, and two-sample (independent-sample) t -tests. The main emphasis is on identifying hypotheses and solving numerical problems step by step.

The two-sample chapter covers both the pooled-variance and Welch methods, with hypotheses, assumptions, worked calculations, graphs, and interpretation. The notes focus throughout on inference about means using t -tests.

How to study. Read the foundations first. Work through the illustrations with a calculator. Then attempt the practice questions before checking the answers. Use the formula sheet and final checklist for revision. For two-sample testing, begin with Section 12 on page 40; its practice questions include Q25–Q30 and Q37–Q40.

Printing. All text, tables, curves, arrows, and hatching are black on white. No diagram requires colour. Illustrations use invented teaching data.

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1 Hypothesis Testing: Foundations

1.1 What Is Hypothesis Testing?

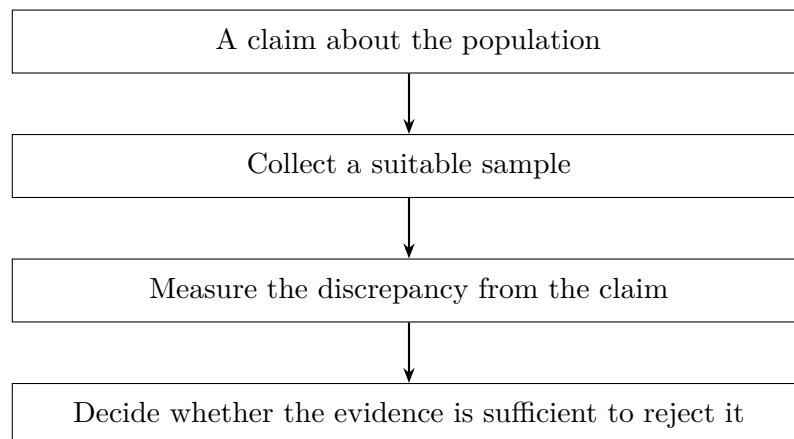
Hypothesis testing provides a systematic way to examine a claim about a population using information collected from a sample.

Examples include:

- Does a filling machine deliver a mean of 500 ml?
- Has a training programme increased the mean examination score?
- Does a new procedure reduce the mean time needed for a task?
- Do two machines produce items with different mean lengths?

We usually cannot measure every member of the population. Instead, we observe a sample and ask whether its result is reasonably consistent with a stated claim.

Fundamental idea
Assume the null hypothesis is true. Ask how unusual the sample result would be under that assumption. If it is sufficiently unusual in the direction specified by the alternative hypothesis, reject the null hypothesis.

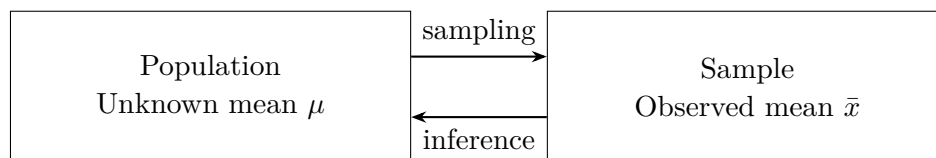


A test does not remove uncertainty. It gives a decision rule with a controlled chance of a particular kind of error, provided the model assumptions hold.

1.2 Population, Sample, and Statistical Inference

A **population** is the complete collection about which we want information. A **sample** is the collection actually observed.

Suppose a factory produces thousands of bottles each day. All bottles produced under the process of interest form the population. Ten bottles selected for measurement form a sample.



Statistical inference means using sample information to draw conclusions about a population. Two common tasks are estimation and hypothesis testing.

Task	Question	Example
Estimation	What values are plausible?	Estimate the population mean filling volume.
Testing	Is there sufficient evidence against a specified claim?	Test whether the mean equals 500 ml.

The sample mean varies from sample to sample. Therefore, $\bar{x} = 498$ does not automatically contradict $\mu = 500$. A difference of 2 ml may be ordinary sampling variation, or it may be unusually large. The sample size and variability help us decide which interpretation is reasonable.

The test concerns the *population mean*, even though the calculation uses the *sample mean*. Keep these two quantities separate throughout a solution.

1.3 Parameters, Statistics, and Notation

A **parameter** describes a population. A **statistic** is calculated from sample observations.

Quantity	Population	Sample
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Variance	σ^2	s^2
Size	Often large or unspecified	n

For observations $x_1, x_2, \dots, x_n$,

$$\bar{x} = \frac{\sum x_i}{n}, \quad s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}, \quad s = \sqrt{s^2}.$$

The symbol μ_0 denotes the mean specified by the null hypothesis. It is a fixed reference value supplied by the question.

Illustration – Identify the quantities

A company claims that its batteries last 40 hours on average. A sample of 12 batteries gives a mean of 38.5 hours and a sample standard deviation of 3 hours.

$$\mu_0 = 40, \quad n = 12, \quad \bar{x} = 38.5, \quad s = 3.$$

The actual population mean μ is unknown. The number 38.5 is an observed sample result; it is not the population claim.

Units matter. The mean and standard deviation have the units of the observations. Variance has squared units. The final t statistic has no units, because its numerator and denominator have the same units.

In the following notes, H_1 and H_a mean the same thing: the alternative hypothesis. We use H_1 consistently.

2 Null and Alternative Hypotheses

2.1 The Two Competing Statements

Null hypothesis

The **null hypothesis**, denoted by H_0 , is the statement against which the sample evidence is assessed. In a basic mean test it specifies a reference mean, or includes a reference mean as the boundary of an inequality.

Alternative hypothesis

The **alternative hypothesis**, denoted by H_1 , describes the departure from H_0 for which we seek evidence. It determines the direction of the test.

For a claim about a mean of 50, the possible pairs are:

$$\begin{aligned} H_0 : \mu = 50 & \quad \text{against} \quad H_1 : \mu \neq 50, \\ H_0 : \mu \leq 50 & \quad \text{against} \quad H_1 : \mu > 50, \\ H_0 : \mu \geq 50 & \quad \text{against} \quad H_1 : \mu < 50. \end{aligned}$$

For a one-sided test, calculations use the boundary value $\mu_0 = 50$. Many introductory solutions therefore write $H_0 : \mu = 50$ as shorthand even in the one-sided case. The full null includes all values opposite to the alternative.

Equality belongs in the null. The alternative uses $>$, $<$, or $\neq$. However, it is not correct to say that the null always means “no effect”. A null may specify any reference mean or reference difference required by a problem.

The claim is not always H_0 . A claim that the mean has increased is usually the alternative. A claim that the mean equals a stated value is normally the null. Identify the mathematical statement before assigning its label.

2.2 How to Identify the Hypotheses

SOP – Translate words into hypotheses

1. Identify the population and the measured variable.
2. Define the parameter: μ , μ_d , or $\mu_1 - \mu_2$.
3. Identify the reference value: μ_0 or Δ_0 .
4. Ask what departure the question wants evidence for.
5. Write that departure as H_1 using $>$, $<$, or $\neq$.
6. Write the complementary null, including equality.
7. Select the tail from H_1 , before examining the sample result.

Illustration. A machine is set to fill 500 ml. An inspector suspects that it is underfilling bottles.

Let μ be the population mean filling volume. Underfilling means a mean below 500 ml. Therefore,

$$H_0 : \mu \geq 500, \quad H_1 : \mu < 500.$$

The test is left-tailed, and the calculation uses $\mu_0 = 500$.

Flipped question. Suppose the inspector instead asks whether the setting has changed in either direction. Then

$$H_0 : \mu = 500, \quad H_1 : \mu \neq 500.$$

This is two-tailed. The reference value is unchanged, but the question is different.

Do not use the observed mean to choose the alternative. Even if the sample happens to have $\bar{x} < 500$, a pre-specified two-sided question remains two-sided. The direction comes from the research question, not from a convenient feature of the observed data.

2.3 A Translation Chart

Question or departure being investigated	H_1	Tail
Is the mean different from k ?	$\mu \neq k$	Two
Has the mean changed from k ?	$\mu \neq k$	Two
Has the mean increased above k ?	$\mu > k$	Right
Is the mean more than k ?	$\mu > k$	Right
Has the mean fallen below k ?	$\mu < k$	Left
Is the mean less than k ?	$\mu < k$	Left

Words such as “better”, “improved”, and “effective” require interpretation.

- For marks, better usually means a *higher* mean score.
- For completion time, better usually means a *lower* mean time.
- For manufacturing, better might mean closer to a target, which requires a more precise question before a direction can be selected.

Illustration – The same word, different tails

A course is intended to improve marks from a benchmark of 60:

$$H_1 : \mu > 60.$$

A procedure is intended to improve delivery time from a benchmark of 30 minutes:

$$H_1 : \mu < 30.$$

The word “improve” does not itself tell us whether to use $>$ or $<$.

Also distinguish a statement about the *mean* from a statement about every individual. “The mean battery life is at least 10 hours” does not say that every battery lasts at least 10 hours. A mean t -test addresses the former.

2.4 Flipped Claims: At Least and At Most

These expressions are easiest to handle by writing the claim first, then asking whether the problem seeks to challenge it or to establish a strict departure.

Illustration A – Challenge an “at least” claim

A supplier claims a population mean life of at least 1000 hours. A customer suspects that the claim is overstated.

$$H_0 : \mu \geq 1000, \quad H_1 : \mu < 1000.$$

The supplier's claim belongs in H_0 . Evidence of shorter life leads to rejection.

Illustration B – Challenge an “at most” claim

A service claims a population mean waiting time of at most 10 minutes. A customer suspects that the mean is actually longer.

$$H_0 : \mu \leq 10, \quad H_1 : \mu > 10.$$

This time the challenge is right-tailed.

Flipped aim. Suppose a new service wants evidence that its mean waiting time is *below* 10 minutes. Then

$$H_0 : \mu \geq 10, \quad H_1 : \mu < 10.$$

Failing to reject the earlier claim $\mu \leq 10$ would not establish this new claim. “There is insufficient evidence that the service is too slow” and “there is evidence that the service is faster than the benchmark” are different conclusions.

Practical rule. Read the sentence containing “test whether”, “suspects”, or “find evidence that”. It normally identifies the question that H_1 must answer.

3 One-Tailed and Two-Tailed Tests

3.1 The Alternative Determines the Rejection Region

A **rejection region** contains values of the test statistic regarded as sufficiently inconsistent with H_0 in the direction specified by H_1 .

For a one-sample t -test,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

Because the denominator is positive, the sign of t is the sign of $\bar{x} - \mu_0$.

Alternative	Evidence against H_0	Rejection rule
$\mu > \mu_0$	Sufficiently large positive t	$t > c$
$\mu < \mu_0$	Sufficiently large negative t	$t < -c$
$\mu \neq \mu_0$	Sufficiently large magnitude in either direction	$ t > c$

Here c is a positive critical value. It depends on the significance level, the degrees of freedom, and whether the test has one or two tails.

For a two-sided 5% test, split the total rejection probability equally:

$$0.05 = 0.025 + 0.025.$$

For a one-sided 5% test, put the entire 0.05 in the specified tail.

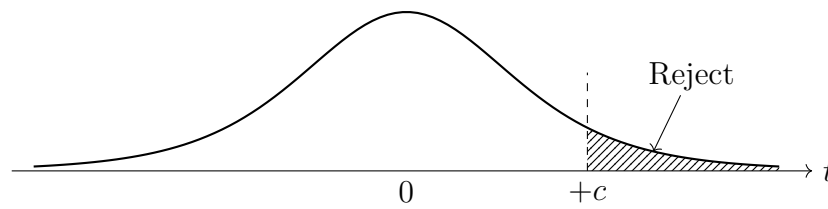
Important distinction

Taking $|t|$ is appropriate for the usual symmetric two-sided test. It is not a way to remove an inconvenient negative sign in a one-sided test.

If $H_1 : \mu > \mu_0$ but $t = -3$, the sample points in the opposite direction. It does not provide evidence for an increase.

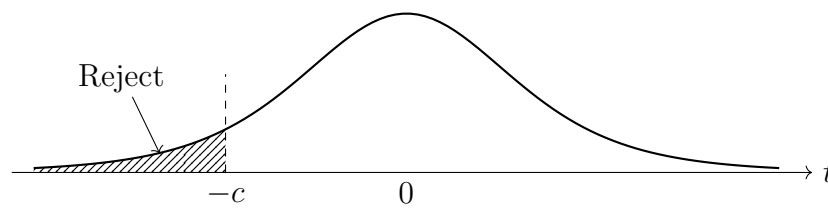
3.2 Visualisation of the Three Tests

Right-tailed: $H_1 : \mu > \mu_0$. Reject beyond the positive cutoff.



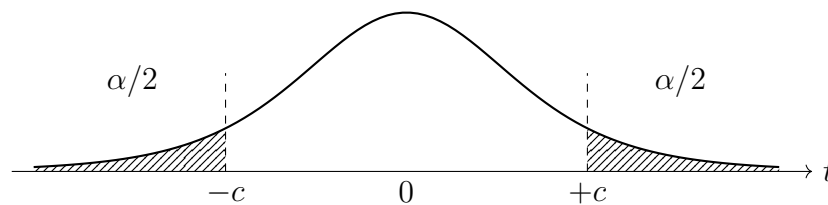
One rejection tail with total area α .

Left-tailed: $H_1 : \mu < \mu_0$. Reject below the negative cutoff.



One rejection tail with total area α .

Two-tailed: $H_1 : \mu \neq \mu_0$. Reject beyond either cutoff.



Two rejection tails, each with area $\alpha/2$.

These are schematic diagrams. Hatching identifies the rejection regions; the exact cutoff is obtained from the appropriate t table.

4 Significance, Errors, and Decisions

4.1 Type I and Type II Errors

A statistical decision can be correct or incorrect because it is made from a sample.

Decision	H_0 is true	H_0 is false
Reject H_0	Type I error	Correct rejection
Do not reject H_0	Correct non-rejection	Type II error

A **Type I error** means rejecting a true null hypothesis. The significance level α controls this error probability. For a one-sided composite null, the rejection probability is at most α under H_0 and equals α at the boundary in the continuous exact model.

A **Type II error** means failing to reject a false null hypothesis. Its probability is denoted by β and depends on the actual alternative value, the sample size, and the variability.

$$\text{Power} = 1 - \beta.$$

Power is the probability of rejecting H_0 at a specified alternative value.

Illustration – Mean filling volume

For $H_0 : \mu = 500$ against $H_1 : \mu \neq 500$:

- Type I error: conclude that the mean differs from 500 ml when it is 500 ml.
- Type II error: fail to detect a departure when the true mean differs from 500 ml.

A smaller α makes rejection more demanding. With the same design and sample size, this generally reduces power against a given departure. It does not automatically make every aspect of the test better.

4.2 Significance Level and the p-Value

The **significance level** is chosen before looking at the results. Common choices are 0.05 and 0.01, corresponding to 5% and 1%.

p-value

Under the null model, the p-value is the probability of obtaining a test statistic at least as extreme as the observed value, where “extreme” is defined by H_1 . For a one-sided mean test, calculate it at the null boundary.

Let T have the relevant t distribution under the null boundary and let t_{obs} be the observed statistic.

$$\begin{aligned} H_1 : \mu > \mu_0 : \quad p &= P(T \geq t_{\text{obs}}), \\ H_1 : \mu < \mu_0 : \quad p &= P(T \leq t_{\text{obs}}), \\ H_1 : \mu \neq \mu_0 : \quad p &= 2P(T \geq |t_{\text{obs}}|). \end{aligned}$$

The last equality uses symmetry of the t distribution.

Decision rule:

$$p \leq \alpha \implies \text{reject } H_0; \quad p > \alpha \implies \text{do not reject } H_0.$$

Illustration. A correctly calculated p-value is 0.032. At 5%, reject because $0.032 < 0.05$. At 1%, do not reject because $0.032 > 0.01$.

The p-value is *not* $P(H_0 \text{ is true})$. It is also not the probability that the result happened “by chance”. It measures extremeness under a specified null model and its assumptions.

4.3 How to Write the Final Decision

Use one of the following forms.

When the result is significant

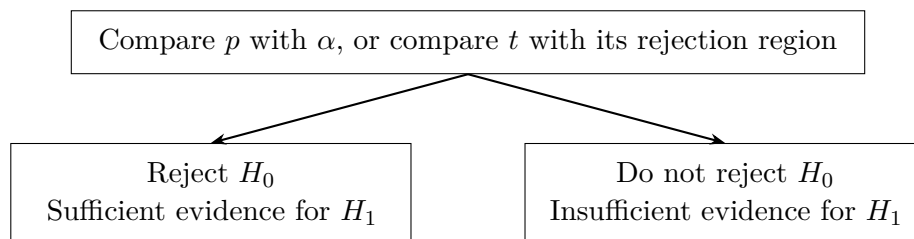
“At the 5% significance level, reject H_0 . There is sufficient evidence that the population mean filling volume is below 500 ml.”

When the result is not significant

“At the 5% significance level, do not reject H_0 . There is insufficient evidence that the population mean filling volume is below 500 ml.”

The second statement does not prove that the mean equals 500 ml, nor does it prove that the process is satisfactory. A sample can be too small or too variable to detect a meaningful departure.

Some examination solutions say “accept H_0 ”. The more precise wording is “do not reject H_0 ” or “fail to reject H_0 ”. If the former wording is used, understand it only as a decision at the chosen significance level, not as proof.



Always identify the population, variable, direction, and level in the conclusion. “Significant” by itself does not answer the original question.

5 Why We Use a t -Test

5.1 Unknown Population Standard Deviation

A mean test needs to compare an observed difference with the uncertainty in the sample mean. In many practical problems the population standard deviation σ is unknown.

We estimate it using the sample standard deviation s . This introduces extra uncertainty. Student's t distribution accounts for that extra uncertainty in the normal-population model.

One-sample t statistic

For a random sample from a normal population with unknown variance,

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \quad \text{when } \mu = \mu_0.$$

The observed statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

The numerator measures the observed departure from the null mean. The denominator estimates how much a sample mean typically varies from sample to sample.

A useful interpretation:

$$t = \frac{\text{observed difference from the null}}{\text{estimated standard error}}.$$

If $t = 2$, the sample mean is two estimated standard errors above the null mean. If $t = -2$, it is two estimated standard errors below it.

The t -test is often introduced under “small samples”, but it is not restricted to $n < 30$. With unknown σ , it also applies to larger samples under suitable conditions. There is no universal sample-size switch that fixes every assumption.

5.2 Standard Deviation versus Standard Error

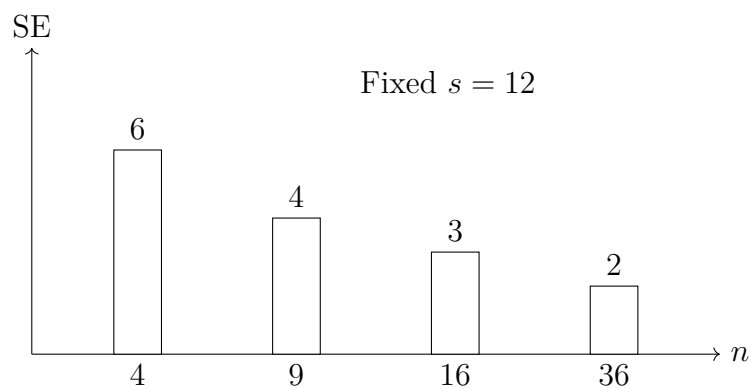
These two quantities answer different questions.

Quantity	Formula	What it describes
Sample standard deviation	s	Spread of individual observations
Estimated standard error	$s/\sqrt{n}$	Uncertainty in the sample mean

If $s = 12$ and $n = 36$, then

$$\text{SE}(\bar{x}) = \frac{12}{\sqrt{36}} = 2.$$

Individual values vary on a scale of about 12 units. The estimated sampling variation of the mean is on a scale of 2 units.



Why divide by $\sqrt{n}$? For independent observations with variance σ^2 ,

$$\text{Var}(\bar{X}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Taking the square root gives $\sigma/\sqrt{n}$. Replacing unknown σ by s gives the estimated standard error used in the t statistic.

5.3 Degrees of Freedom

For one sample, the degrees of freedom are

$$\nu = n - 1.$$

The symbol ν is the Greek letter nu. Tables may use “df” or “d.f.” instead.

Once the sample mean is calculated, the deviations satisfy

$$\sum_{i=1}^n (x_i - \bar{x}) = 0.$$

Therefore, only $n - 1$ deviations can vary freely; the final deviation is determined by the others. This is the basic reason for the $n - 1$ in the sample variance and in the one-sample t distribution.

Illustration – One constraint

Suppose three observations have mean 10. Their sum must be 30. If the first two observations are 8 and 11, the third must be 11.

After fixing the mean, two observations can be chosen freely. The third cannot. Thus the corresponding number of free deviations is $3 - 1 = 2$.

Different designs use different degrees of freedom:

Design	Degrees of freedom
One sample of n observations	$n - 1$
Paired sample of n complete pairs	$n - 1$
Two independent samples, pooled variance	$n_1 + n_2 - 2$
Welch independent-sample test	Calculated approximation, often non-integer

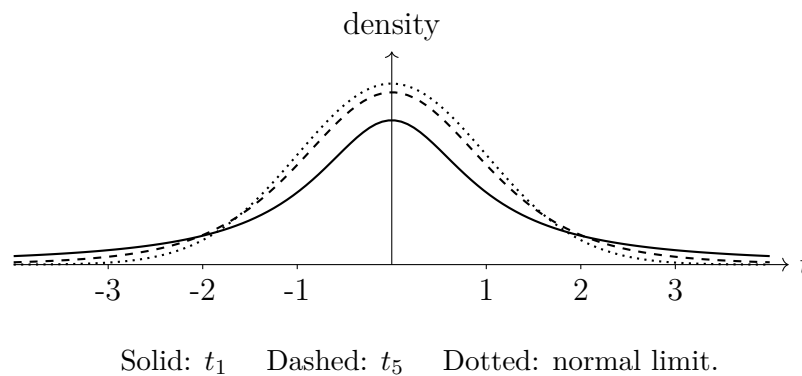
Degrees of freedom select the correct reference distribution. They are not an optional detail that can be supplied after choosing a convenient critical value.

6 Student's t Distribution

6.1 Shape and Properties

The t distribution is continuous, symmetric about 0, and bell-shaped. Its total area is 1. Its shape depends on the degrees of freedom.

- With fewer degrees of freedom, the tails are heavier.
- With more degrees of freedom, the curve approaches the standard normal curve.
- Symmetry means the negative critical value is the negative of the positive one.
- There is a family of t curves, not one single curve for all sample sizes.



Heavier tails mean that a larger magnitude can be needed before the observation is considered unusual. This reflects uncertainty from estimating variability.

For example, at the two-sided 5% level, the positive critical value is 2.776 for $\nu = 4$, but 2.045 for $\nu = 29$.

Careful wording. Every t curve is centred symmetrically at 0, but its mean exists only for $\nu > 1$. This detail does not change the tail-based testing rules used in the notes.

6.2 Assumptions and Conditions

Before using a formula, check what the observations represent and how they were collected.

1. **Quantitative response.** The test concerns a numerical mean, such as length, time, volume, or score.
2. **Suitable sampling.** Random sampling or a defensible study design supports inference to the population of interest.
3. **Independence.** In a one-sample test, observations should be independent. In a paired test, different pairs should be independent.
4. **Distributional condition.** Exact one-sample and paired t inference uses a normal model for the observations or the pair differences, respectively.
5. **Nonzero variability.** The estimated standard error must be positive.

For small samples, severe skewness or extreme outliers can make t inference unreliable. Larger samples often allow useful approximations, but a fixed rule such as “ $n \geq 30$ means everything is valid” is too strong.

What changes between designs?

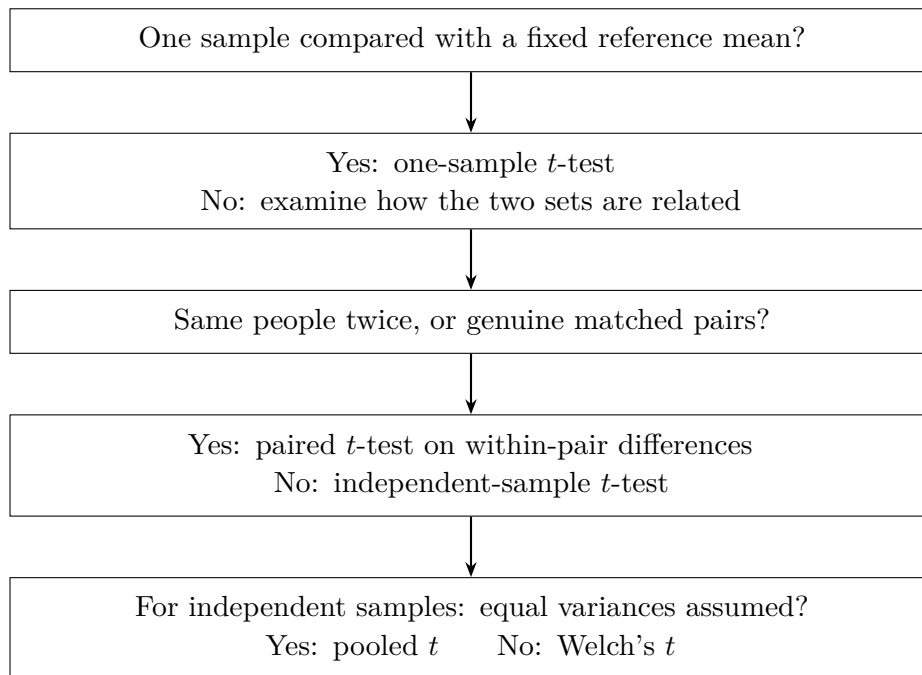
Paired: normality concerns the distribution of the differences, not necessarily each set of before and after measurements separately.

Pooled independent samples: additionally assume equal population variances.

Welch: does not assume equal population variances; it still needs suitable sampling, independence, and appropriate distributional conditions.

In numerical examination questions, state the assumptions supplied by the question. If an assumption is needed but not stated, say that the conclusion is conditional on it. A correct calculation cannot repair an unsuitable study design.

6.3 Choosing the Correct t -Test



The arrows are a reading sequence; follow only the branch matching the design.

Illustrations:

- Ten bulbs compared with a claimed mean life of 1000 hours: one sample.
- The same ten students tested before and after coaching: paired.
- Ten students taught by method A and twelve different students taught by method B: independent samples.

Two samples do not become paired merely because they have the same size. Pairing must come from the study design. Also, a second sample mean is not a known population benchmark; its uncertainty must be included in the comparison.

7 Using the t Table

7.1 Our Critical-Value Convention

We use $t_{q,\nu}$ for the **positive** critical value leaving right-tail area q :

$$P(T_\nu > t_{q,\nu}) = q.$$

This is an upper-tail convention. Some books instead use a cumulative-probability subscript; their $t_{1-q,\nu}$ represents our $t_{q,\nu}$.

SOP – Read the table

1. Calculate the degrees of freedom ν .
2. Decide whether the question is one-tailed or two-tailed.
3. For a one-tailed level α , use column $q = \alpha$.
4. For a two-tailed level α , use column $q = \alpha/2$.
5. Read the positive value at the row and column intersection.
6. For a left-tailed test, attach a minus sign to the cutoff.

Illustration. With $n = 10$, we have $\nu = 9$.

Right-tailed 5% : $t > t_{0.05,9} = 1.833$,

Left-tailed 5% : $t < -1.833$,

Two-tailed 5% : $|t| > t_{0.025,9} = 2.262$.

The two-tailed cutoff is larger because only half of α is placed in each tail.

Keep full calculator precision until the final comparison. If a rounded statistic appears exactly equal to a rounded cutoff, use more precision rather than treating rounded equality as a decisive result.

7.2 Selected Critical Values

Every entry is positive and leaves area q to its right. Values are rounded to three decimal places. In a two-sided test, remember to use $q = \alpha/2$.

ν	$q = 0.10$	$q = 0.05$	$q = 0.025$	$q = 0.005$
1	3.078	6.314	12.706	63.657
2	1.886	2.920	4.303	9.925
3	1.638	2.353	3.182	5.841
4	1.533	2.132	2.776	4.604
5	1.476	2.015	2.571	4.032
6	1.440	1.943	2.447	3.707
7	1.415	1.895	2.365	3.499
8	1.397	1.860	2.306	3.355
9	1.383	1.833	2.262	3.250
10	1.372	1.812	2.228	3.169
11	1.363	1.796	2.201	3.106
12	1.356	1.782	2.179	3.055
13	1.350	1.771	2.160	3.012
14	1.345	1.761	2.145	2.977
15	1.341	1.753	2.131	2.947
16	1.337	1.746	2.120	2.921
17	1.333	1.740	2.110	2.898
18	1.330	1.734	2.101	2.878
19	1.328	1.729	2.093	2.861
20	1.325	1.725	2.086	2.845
24	1.318	1.711	2.064	2.797
25	1.316	1.708	2.060	2.787
29	1.311	1.699	2.045	2.756
30	1.310	1.697	2.042	2.750
40	1.303	1.684	2.021	2.704
60	1.296	1.671	2.000	2.660
120	1.289	1.658	1.980	2.617

Values can be checked against the NIST Student's t critical-value table; NIST labels columns by cumulative probability $1 - q$.

If the required row is absent, use an appropriate fuller table or software. A smaller available df gives a conservative, larger cutoff; identify this choice explicitly. Do not substitute an unrelated row silently.

8 The General t -Test Procedure

SOP – A complete examination solution

1. **Identify the design:** one sample, paired, or independent samples.
2. **Define the parameter:** say what mean or difference is being tested.
3. **State H_0 and H_1 :** translate the question carefully.
4. **State α :** normally supplied in the question.
5. **Check conditions:** record the relevant assumptions.
6. **Calculate the standard error and t :** show the formula before substitution.
7. **Find degrees of freedom and the cutoff:** match the correct tail.
8. **Make the comparison:** show the numerical inequality or p-value.
9. **Conclude in words:** answer the actual population question.

The critical-value and p-value approaches express the same decision for the same test and significance level. A solution may use either, unless the question specifies a method.

A compact working template:

Hypotheses $\rightarrow$ SE $\rightarrow t \rightarrow$ df $\rightarrow$ cutoff $\rightarrow$ conclusion
--

Before calculation, predict the sign. If $\bar{x} < \mu_0$, a one-sample t value must be negative. This simple check catches many substitution errors.

During calculation, preserve precision. Round the final t value to three decimal places when reporting it, but use more digits for close decisions.

After calculation, check the wording. A two-sided result supports a difference. A one-sided result supports the specified direction. Neither says that every individual measurement changed.

9 One-Sample t -Test

9.1 Formula and Interpretation

Use a one-sample t -test when one numerical sample is compared with a fixed population-mean benchmark and the population standard deviation is unknown.

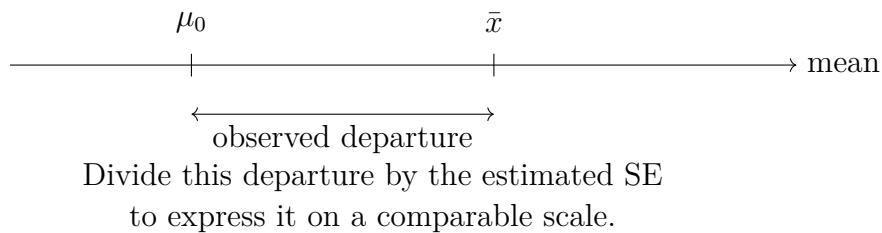
One-sample formula

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad \nu = n - 1.$$

Here $\bar{x}$ is the sample mean, μ_0 is the reference mean, s is the sample standard deviation, and n is the sample size.

Consider two samples with the same observed difference $\bar{x} - \mu_0 = 4$. If the estimated standard error is 1, then $t = 4$. If the estimated standard error is 4, then $t = 1$.

The same raw difference can therefore provide very different evidence. A large difference relative to its uncertainty is more surprising under the null model than the same difference measured with high uncertainty.



Under the normal model, replacing σ with s produces the t_{n-1} reference distribution. A derivation of that distribution is not needed to solve the elementary testing problems here.

Check the data description. If the question gives a variance of 25, then $s = 5$, not 25. If it gives the standard error directly, do not divide that quantity by $\sqrt{n}$ again.

9.2 Illustration: A Two-Tailed Test

Illustration – Is the filling mean different?

A random sample of 10 bottles has mean volume 496 ml and sample standard deviation 5 ml. Assuming a normal population, test whether the mean filling volume differs from 500 ml at the 5% level.

Let μ be the population mean volume.

$$H_0 : \mu = 500, \quad H_1 : \mu \neq 500, \quad \alpha = 0.05.$$

1. Calculate the estimated standard error

$$SE = \frac{s}{\sqrt{n}} = \frac{5}{\sqrt{10}} = 1.5811 \text{ ml.}$$

2. Calculate the test statistic

$$t = \frac{496 - 500}{1.5811} = -2.5298 \approx -2.530.$$

3. Find the rejection region

$$\nu = 10 - 1 = 9, \quad t_{0.025,9} = 2.262.$$

Reject if $t < -2.262$ or $t > 2.262$, equivalently if $|t| > 2.262$.

4. Compare and conclude

$$|t| = 2.530 > 2.262.$$

Reject H_0 . At the 5% level, there is sufficient evidence that the population mean volume differs from 500 ml. The observed departure is downward.

The two-sided p-value is approximately 0.0322, giving the same decision. The calculation concerns the mean of the filling process; it does not imply that all bottles contain less than 500 ml.

9.3 Illustration: A Right-Tailed Test

Illustration – Evidence of an increase

A process previously had a benchmark mean score of 50. A random sample of 16 observations from the new process gives $\bar{x} = 53$ and $s = 4$. Test for an increase at 5%, assuming independent normal observations.

Let μ be the mean score under the new process.

$$H_0 : \mu \leq 50, \quad H_1 : \mu > 50.$$

Use the boundary $\mu_0 = 50$:

$$SE = \frac{4}{\sqrt{16}} = 1, \quad t = \frac{53 - 50}{1} = 3.$$

The degrees of freedom are

$$\nu = 16 - 1 = 15.$$

For a right-tailed 5% test,

$$c = t_{0.05,15} = 1.753.$$

Since

$$3 > 1.753,$$

reject H_0 . There is sufficient evidence that the new process has a population mean score above 50.

The one-sided p-value is approximately 0.00449. This is also below 0.01, so the result would be significant at the 1% level.

Why only one tail? The question asks for evidence of an increase. A large negative t value would indicate a decrease and would not support this alternative. The sign is part of the evidence, not just a calculation detail.

9.4 Illustration: A Left-Tailed Test

Return to the bottle sample:

$$n = 10, \quad \bar{x} = 496, \quad s = 5.$$

This time the inspector's pre-specified question is whether the machine is *underfilling* relative to 500 ml.

$$H_0 : \mu \geq 500, \quad H_1 : \mu < 500.$$

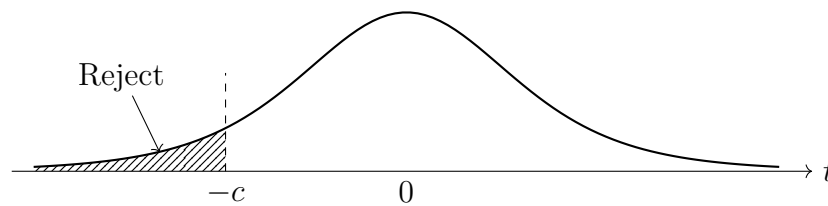
The numerical statistic is unchanged:

$$t = \frac{496 - 500}{5/\sqrt{10}} = -2.530, \quad \nu = 9.$$

The rejection rule is now

$$t < -t_{0.05,9} = -1.833.$$

Since $-2.530 < -1.833$, reject H_0 . There is sufficient evidence at 5% that the population mean filling volume is below 500 ml.



The observed statistic is in the left rejection region.

The left-sided p-value is approximately 0.0161. For this symmetric distribution and an observation in the specified direction, it is half the two-sided value 0.0322.

Do not generalise the halving rule carelessly. If the alternative had been $\mu > 500$, the same negative statistic would give a right-sided p-value of about 0.9839, not 0.0161. Tail direction must be checked first.

9.5 Illustration: Insufficient Evidence

Illustration – A sample difference need not be significant

A sample of 12 independent normal observations has $\bar{x} = 51.2$ and $s = 4$. Test $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$ at 5%.

$$SE = \frac{4}{\sqrt{12}} = 1.1547.$$

Therefore,

$$t = \frac{51.2 - 50}{1.1547} = 1.0392, \quad \nu = 11.$$

The two-sided critical value is

$$t_{0.025,11} = 2.201.$$

Since

$$|1.0392| < 2.201,$$

do not reject H_0 . At the 5% level, the sample does not provide sufficient evidence that the population mean differs from 50.

The p-value is approximately 0.3210. The observed result is not particularly unusual under the null model.

What this conclusion does and does not say

We observed a sample mean above 50. That is a descriptive fact.

The test did not find enough evidence of a population departure. That is an inferential conclusion.

Neither statement establishes that the population mean is exactly 50.

Always distinguish “no observed difference”, “no statistically significant difference”, and “proof of no meaningful difference”. These are not equivalent claims.

9.6 Illustration: Starting from Raw Data

Eight independent observations from an approximately normal population are

$$48, 50, 51, 49, 52, 53, 50, 55.$$

Test whether the population mean differs from 50 at 5%.

$$H_0 : \mu = 50, \quad H_1 : \mu \neq 50.$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
48	-3	9
50	-1	1
51	0	0
49	-2	4
52	1	1
53	2	4
50	-1	1
55	4	16
$\sum x_i = 408$	$\sum (x_i - \bar{x}) = 0$	Sum = 36

Thus

$$\bar{x} = \frac{408}{8} = 51, \quad s^2 = \frac{36}{7} = 5.1429, \quad s = 2.2678.$$

$$\text{SE} = \sqrt{\frac{36}{7 \cdot 8}} = 0.8018, \quad t = \frac{51 - 50}{0.8018} = 1.2472.$$

Here $\nu = 7$, and $t_{0.025,7} = 2.365$. Since $1.2472 < 2.365$, do not reject H_0 . There is insufficient evidence of a mean different from 50.

Arithmetic checks: the deviations add to zero, all squared deviations are nonnegative, and the sign of t agrees with $51 - 50 > 0$.

9.7 Shortcut Variance and Summary Data

When the question gives n , $\sum x_i$, and $\sum x_i^2$, use

$$\bar{x} = \frac{\sum x_i}{n}, \quad s^2 = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1}.$$

The identity behind this formula is

$$\begin{aligned} \sum (x_i - \bar{x})^2 &= \sum x_i^2 - 2\bar{x} \sum x_i + n\bar{x}^2 \\ &= \sum x_i^2 - n\bar{x}^2. \end{aligned}$$

Illustration – Reconstruct the previous calculation

For the eight observations on the previous page,

$$n = 8, \quad \sum x_i = 408, \quad \sum x_i^2 = 20844.$$

Therefore,

$$\bar{x} = 51, \quad s^2 = \frac{20844 - 408^2/8}{7} = \frac{36}{7}.$$

This gives the same $t = 1.2472$ as the deviation table.

If the divisor is different. A question may explicitly give a descriptive variance v calculated with divisor n :

$$v = \frac{\sum (x_i - \bar{x})^2}{n}.$$

Convert it to the usual sample variance before the t calculation:

$$s^2 = \frac{n}{n-1}v.$$

Do not make this correction when the quantity is already the ordinary sample variance with divisor $n-1$. Read the definition given in the question.

10 Flipped One-Sample Examples

10.1 Same Data, Different Research Questions

Suppose a suitable sample gives

$$n = 10, \quad \bar{x} = 52, \quad s = \sqrt{10}, \quad \mu_0 = 50.$$

Then $SE = 1$, $t = 2$, and $\nu = 9$. Compare three questions planned before sampling.

Alternative	Rule at 5%	p-value	Decision
$\mu > 50$	$t > 1.833$	0.0383	Reject
$\mu < 50$	$t < -1.833$	0.9617	Do not reject
$\mu \neq 50$	$ t > 2.262$	0.0766	Do not reject

Right-sided question: the result provides sufficient evidence of an increase, because $2 > 1.833$.

Left-sided question: the observed difference points in the wrong direction. A positive statistic cannot fall in the negative rejection tail.

Two-sided question: the magnitude is not large enough to pass the two-sided cutoff. The available error probability is divided between both tails.

Why this is not a contradiction

The three tests answer different pre-specified questions. Their rejection regions are different. It is therefore possible for the same statistic to pass one rule and fail another.

This comparison is a learning exercise, not permission to try alternatives after seeing the result and report whichever happens to be significant. Selecting a favourable direction after examining the data invalidates the stated error control.

10.2 Same Question, Reversed Departure

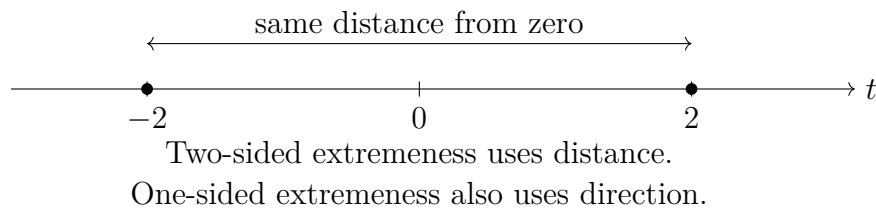
Keep $n = 10$, $s = \sqrt{10}$, and $\mu_0 = 50$, but change the sample mean from 52 to 48.

$$t = \frac{48 - 50}{1} = -2.$$

The sign changes, while the magnitude stays the same.

Alternative	$t = +2$	$t = -2$	Reason
$\mu > 50$	Reject	Do not reject	Right tail only
$\mu < 50$	Do not reject	Reject	Left tail only
$\mu \neq 50$	Do not reject	Do not reject	Same magnitude

All decisions above use $\nu = 9$ and $\alpha = 0.05$.



For a two-sided test under a symmetric distribution, $+2$ and -2 have the same p-value. For a right-sided test, their p-values are approximately 0.0383 and 0.9617, respectively.

Common trap. If you remove the minus sign before a left- or right-tailed comparison, you may reverse the scientific meaning of the result. Preserve the signed statistic until the decision rule explicitly asks for its absolute value.

10.3 Same Data, Different Significance Levels

Return to the two-sided bottle test with $t = -2.530$ and $\nu = 9$.

Level	Column q	Positive cutoff	Decision
10%	0.05	1.833	Reject
5%	0.025	2.262	Reject
1%	0.005	3.250	Do not reject

The p-value 0.0322 is unchanged because the data, model, and alternative are unchanged. Only the chosen decision threshold changes.

Interpretation. The evidence is sufficient under a 5% rule but not under the more demanding 1% rule. This is not evidence that the calculation has become inconsistent.

Chosen threshold versus observed evidence

α is a rule chosen in advance.

The p-value is calculated from the observed result under the specified null model.

Compare them only after both roles are clear.

A two-sided test has a p-value between 0.01 and 0.05 when its statistic lies between the corresponding positive cutoffs in magnitude. This lets a printed table give a useful p-value interval without evaluating a distribution function.

For this example,

$$2.262 < 2.530 < 3.250 \implies 0.01 < p < 0.05.$$

The table bounds agree with the more precise value 0.0322.

10.4 Reverse Problem: Find a Sample-Mean Cutoff

Sometimes the question asks what sample mean would lead to rejection, rather than asking for a decision from a supplied mean.

Suppose $n = 16$, $s = 4$, $\mu_0 = 50$, and the alternative is $\mu > 50$ at 5%. The rejection rule is

$$\frac{\bar{x} - 50}{4/\sqrt{16}} > t_{0.05,15} = 1.753.$$

Since $4/\sqrt{16} = 1$,

$$\bar{x} - 50 > 1.753, \quad \boxed{\bar{x} > 51.753}.$$

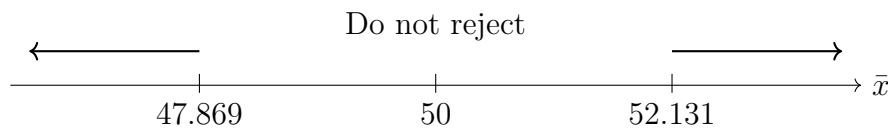
This is a cutoff conditional on the stated sample standard deviation.

Flipped direction: for $H_1 : \mu < 50$,

$$\frac{\bar{x} - 50}{1} < -1.753 \implies \boxed{\bar{x} < 48.247}.$$

Two-sided question: using $t_{0.025,15} = 2.131$,

$$\boxed{\bar{x} < 47.869 \quad \text{or} \quad \bar{x} > 52.131}.$$



General two-sided form:

$$\bar{x} < \mu_0 - c \frac{s}{\sqrt{n}} \quad \text{or} \quad \bar{x} > \mu_0 + c \frac{s}{\sqrt{n}}.$$

Because s is itself a sample quantity, a new sample with a different s has different mean cutoffs. Do not treat these as universal process limits.

11 Paired t -Test

11.1 Why Pairing Changes the Calculation

Paired observations arise when each observation in one set has a meaningful partner in the other set.

- The same student is tested before and after coaching.
- The same product is measured by two instruments.
- Two units are deliberately matched on important characteristics.

The basic observation is now the *within-pair difference*.

Before		After		Difference
40	→	43	→	$43 - 40 = 3$
50	→	54	→	$54 - 50 = 4$
60	→	65	→	$65 - 60 = 5$

Define, for example,

$$d_i = \text{after}_i - \text{before}_i.$$

Then μ_d is the population mean of these differences. Testing no mean change means testing $\mu_d = 0$.

The paired t -test is a one-sample t -test applied to the differences. Individual baseline differences can be removed by subtraction, making the change more precisely estimated when the pairing is informative.

There are n differences for n complete pairs. Twenty before/after measurements on ten people give $n = 10$, not $n = 20$.

11.2 Paired Formula and Hypotheses

For differences $d_1, d_2, \dots, d_n$,

$$\bar{d} = \frac{\sum d_i}{n}, \quad s_d^2 = \frac{\sum (d_i - \bar{d})^2}{n - 1}.$$

The shortcut formula is

$$s_d^2 = \frac{\sum d_i^2 - (\sum d_i)^2/n}{n - 1}.$$

Paired *t* statistic

To test the null-boundary mean difference Δ_0 ,

$$t = \frac{\bar{d} - \Delta_0}{s_d/\sqrt{n}}, \quad \nu = n - 1.$$

For no mean change, set $\Delta_0 = 0$.

If $d = \text{after} - \text{before}$, then:

Question	Alternative
Any mean change?	$H_1 : \mu_d \neq 0$
Mean score increased?	$H_1 : \mu_d > 0$
Mean completion time reduced?	$H_1 : \mu_d < 0$

The standard deviation required is s_d , the spread of the *differences*. It is generally neither $s_{\text{after}} - s_{\text{before}}$ nor the average of the two separate standard deviations.

Normality is assessed for the population of differences. Dependence within a pair is expected; independence between different pairs is the requirement.

11.3 Illustration: Before and After Coaching

Eight students have the following scores. Assume independent students and approximately normal score differences. Test for an increase at 5%.

Student	Before	After	d	d^2
1	40	42	2	4
2	45	49	4	16
3	50	53	3	9
4	55	60	5	25
5	60	64	4	16
6	65	67	2	4
7	70	73	3	9
8	75	80	5	25
Total			28	108

Define $d = \text{after} - \text{before}$. Then

$$H_0 : \mu_d \leq 0, \quad H_1 : \mu_d > 0.$$

$$\bar{d} = \frac{28}{8} = 3.5, \quad s_d^2 = \frac{108 - 28^2/8}{7} = \frac{10}{7} = 1.4286.$$

Thus $s_d = 1.1952$, and

$$\text{SE}(\bar{d}) = \frac{1.1952}{\sqrt{8}} = 0.4226, \quad t = \frac{3.5 - 0}{0.4226} = 8.2825.$$

With $\nu = 7$, the right-tailed cutoff is $t_{0.05,7} = 1.895$. Since $8.2825 > 1.895$, reject H_0 .

There is sufficient evidence of a positive population mean score change. The observed average increase is 3.5 marks.

Interpretation limit. A before/after increase alone does not prove that coaching caused the increase. Other changes over time may also matter; causal claims depend on the study design, not just the significance of the paired test.

11.4 Flipped Pairing: Reverse the Subtraction

Suppose we use the same coaching data but define

$$d' = \text{before} - \text{after} = -d.$$

Then

$$\bar{d}' = -3.5, \quad s_{d'} = 1.1952, \quad t' = -8.2825.$$

The standard deviation does not change because the squared deviations are unchanged:

$$(d'_i - \bar{d}')^2 = (-d_i + \bar{d})^2 = (d_i - \bar{d})^2.$$

An increase in scores now means a negative value of before minus after:

$$H_0 : \mu_{d'} \geq 0, \quad H_1 : \mu_{d'} < 0.$$

The rejection rule is $t' < -1.895$. Since $-8.2825 < -1.895$, reject H_0 . The scientific conclusion is exactly the same.

Definition	Alternative for improvement	Statistic
After – before	$\mu_d > 0$	+8.2825
Before – after	$\mu_{d'} < 0$	–8.2825

The rule for flipped differences

If the subtraction order is reversed, reverse the sign of the mean difference, the sign of the null difference if nonzero, and the direction of a one-sided alternative. The final substantive conclusion must remain unchanged.

Changing the subtraction without changing the alternative is not an equivalent test. It asks the opposite directional question.

11.5 Illustration: A Paired Result That Is Not Significant

Six independent pairs produce the differences

$$d = 3, -1, 2, -2, 4, 0,$$

where $d = \text{before time} - \text{after time}$. Positive differences indicate reduced time. Assume approximately normal differences and test for a reduction at 5%.

$$H_0 : \mu_d \leq 0, \quad H_1 : \mu_d > 0.$$

We have

$$\sum d_i = 6, \quad \sum d_i^2 = 34, \quad n = 6.$$

Therefore,

$$\bar{d} = 1, \quad s_d^2 = \frac{34 - 6^2/6}{5} = \frac{28}{5} = 5.6.$$

$$\text{SE} = \sqrt{\frac{5.6}{6}} = 0.9661, \quad t = \frac{1}{0.9661} = 1.0351.$$

With $\nu = 5$, the right-tailed cutoff is 2.015. Since $1.0351 < 2.015$, do not reject H_0 .

The sample shows an average reduction of one time unit, but it does not provide sufficient evidence of a positive population mean reduction at 5%.

Why might this happen? Some pairs improved while others became slower. The variability of changes is large relative to the mean change. A positive average alone is not sufficient for rejection.

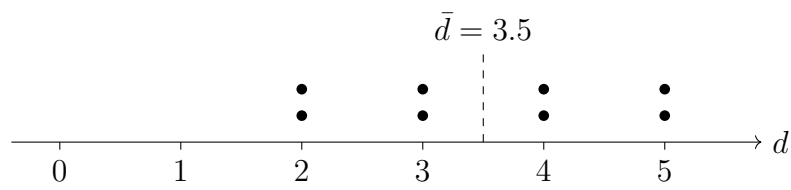
Do not turn the test into a count of how many differences are positive. The paired t -test uses the numerical magnitudes of the differences, their mean, and their estimated standard error.

11.6 Paired-Test SOP and Visual Check

SOP – Paired t -test

1. Verify that the observations form genuine pairs.
2. Write the subtraction order explicitly.
3. Calculate each difference and its square.
4. Find n , $\sum d_i$, and $\sum d_i^2$.
5. State hypotheses about μ_d , using the chosen sign convention.
6. Calculate $\bar{d}$, s_d , and $s_d/\sqrt{n}$.
7. Calculate t and use $\nu = n - 1$.
8. Select the critical region and conclude about the mean change.

For the coaching example, the differences are 2, 4, 3, 5, 4, 2, 3, 5.



This dot plot shows positive differences concentrated around 3.5. It helps us understand the numerical result, but it does not replace the calculation.

Incomplete pairs. A difference requires both measurements. Use the number of complete pairs in a simple paired calculation, and consider whether missing measurements could bias the result. Do not invent a missing partner.

12 Two-Sample t -Test: Independent Population Means

12.1 Comparing Two Population Means

The **two-sample t -test**, also called the **independent-samples** or **unpaired t -test**, compares two population means. Use it when two separate groups supply numerical observations and there is no meaningful pairing between the groups.

Let μ_1 and μ_2 denote the two population means. The parameter of interest is

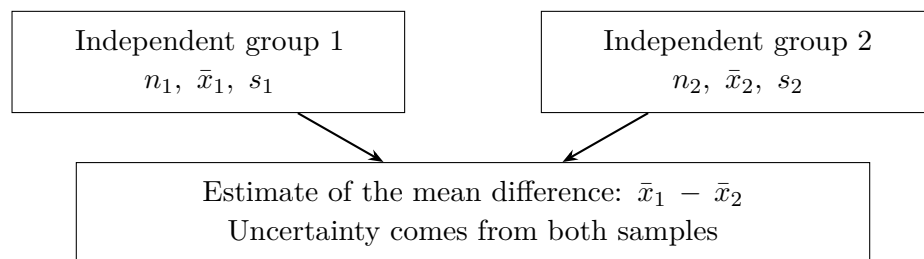
$$\Delta = \mu_1 - \mu_2.$$

For a test of equal means, $\Delta_0 = 0$:

$$H_0 : \mu_1 - \mu_2 = 0, \quad H_1 : \mu_1 - \mu_2 \neq 0.$$

For evidence that group 1 has the higher mean,

$$H_0 : \mu_1 - \mu_2 \leq 0, \quad H_1 : \mu_1 - \mu_2 > 0.$$



Two versions:

- The pooled test assumes a common population variance.
- Welch's test estimates uncertainty separately for the two groups and does not require equal population variances.

Neither version requires the two sample sizes to be equal. Conversely, equal sample sizes do not imply equal population variances or create a paired design.

12.2 Two-Sample Hypotheses and Assumptions

Always define the subtraction order first. Write $\Delta = \mu_1 - \mu_2$ and let Δ_0 be the population difference stated by the null hypothesis. For equal-mean testing, $\Delta_0 = 0$.

Question	H_0	H_1	Tail
Different means?	$\Delta = 0$	$\Delta \neq 0$	Two
Group 1 higher?	$\Delta \leq 0$	$\Delta > 0$	Right
Group 1 lower?	$\Delta \geq 0$	$\Delta < 0$	Left
Difference exceeds k ?	$\Delta \leq k$	$\Delta > k$	Right

For a one-sided test, calculate at the boundary of the null. The sign of the test statistic follows $(\bar{x}_1 - \bar{x}_2) - \Delta_0$, not just the observed mean difference when the benchmark is nonzero.

Conditions for an independent two-sample test

1. The response is numerical, and the target is a difference of population means.
2. Observations are independent within each group and the two groups are independent of each other. Repeated observations on the same person require a paired analysis or another model that accounts for dependence.
3. Sampling or the study design supports inference to the stated populations.
4. For small samples, use appropriate normal-population assumptions for each group and check for severe skewness or extreme outliers. Large samples can support approximate inference under suitable conditions.
5. The pooled test additionally assumes $\sigma_1^2 = \sigma_2^2$. Welch's test does not require this equality and uses an approximate t reference.

Sample size does not select the variance model. Equal sample sizes do not justify pooling. Different sample sizes do not prevent a pooled test when equal population variances are justified.

Two samples versus two tails. “Two-sample” describes the data design; “two-tailed” describes H_1 . A two-sample test can be left-, right-, or two-tailed.

12.3 Pooled Variance: Formula and Reason

If the populations have a common unknown variance, combine the two sample variance estimates using their degrees of freedom as weights:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

Each numerator term is a within-group sum of squared deviations. The denominator is the combined number of free deviations after estimating two group means.

Pooled independent-sample <i>t</i> statistic
$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{s_p \sqrt{1/n_1 + 1/n_2}}, \quad \nu = n_1 + n_2 - 2.$ For equal-mean testing, use $\Delta_0 = 0$.

Because the two sample means are independent,

$$\text{Var}(\bar{X}_1 - \bar{X}_2) = \text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2).$$

Under the common variance assumption, this becomes

$$\sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right).$$

Replacing σ^2 by s_p^2 and taking a square root gives the estimated standard error in the denominator.

Do not average standard deviations. Pool the *variances*, with their df weights, and then take a square root if needed.

Do not infer equal variance merely from similar-looking sample SDs. Use the pooled method when that population assumption is stated or otherwise justified. When it is not justified, Welch's formulation avoids that assumption.

12.4 Illustration: Pooled Two-Sided Test

Two independent normal populations are assumed to have equal variances. Samples give:

Group	n	Mean	Sample SD
1	10	52	4
2	10	48	4

Test whether the population means differ at 5%.

$$H_0 : \mu_1 - \mu_2 = 0, \quad H_1 : \mu_1 - \mu_2 \neq 0.$$

The pooled variance is

$$s_p^2 = \frac{9(16) + 9(16)}{18} = 16, \quad s_p = 4.$$

The estimated standard error is

$$SE = 4\sqrt{\frac{1}{10} + \frac{1}{10}} = \sqrt{3.2} = 1.7889.$$

Therefore,

$$t = \frac{52 - 48}{1.7889} = 2.2361, \quad \nu = 18.$$

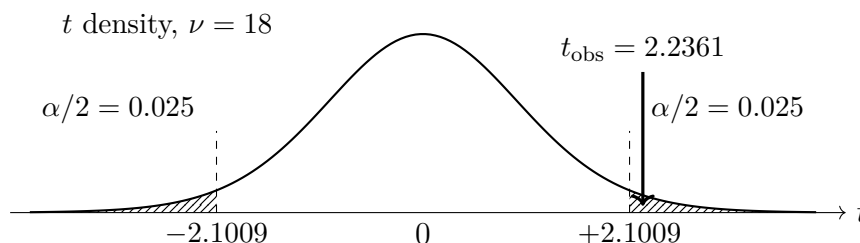
The two-sided 5% critical value is $t_{0.025,18} = 2.101$. Since $2.2361 > 2.101$, reject H_0 .

There is sufficient evidence that the two population means differ. The observed mean difference, group 1 minus group 2, is 4 units.

Why not use a one-sample test against 48? The number 48 is a mean estimated from the second sample. It has uncertainty of its own. Treating it as a fixed population benchmark would omit that uncertainty from the denominator.

12.5 Graph: The Two-Sample Rejection Region

For the preceding pooled example, the null states $\mu_1 - \mu_2 = 0$. Under the normal, equal-variance model, the statistic has a t_{18} distribution when this null is true. At $\alpha = 0.05$, the two cutoffs are ± 2.1009 .



The observed statistic lies in the right rejection tail, so reject the equal-mean null. The density is plotted on a numerical t scale. Each rejection tail continues beyond the displayed range.

The p-value uses the observed statistic as its boundary

The hatching above shows the fixed rejection regions beyond ± 2.1009 . The two-sided p-value instead uses both tails beyond ± 2.2361 :

$$p = 2P(T_{18} \geq 2.2361) \approx 0.0382 < 0.05.$$

These are two ways of reaching the same decision.

The corresponding 95% interval estimates the size of the population difference:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{0.025, 18} \text{SE} = 4 \pm 2.1009(1.7889) \approx (0.242, 7.758).$$

The interval excludes the null value 0. Its width also shows why the estimated difference of 4 should not be treated as an exact population value.

Graph-reading rule. Look at the cutoff, the observed statistic, and the specified tail together. The height of the density at the observed value is not the p-value; the p-value is an area under the reference curve.

12.6 Flipped Groups and Directional Claims

In the preceding example,

$$\bar{x}_1 - \bar{x}_2 = 4, \quad t = 2.2361.$$

For the pre-specified question “Does group 1 have a higher population mean?”, use

$$H_0 : \mu_1 - \mu_2 \leq 0, \quad H_1 : \mu_1 - \mu_2 > 0.$$

At 5%, reject if $t > t_{0.05,18} = 1.734$. The result is significant.

Now reverse the group labels and consistently use group 2 minus group 1:

$$\bar{x}_2 - \bar{x}_1 = -4, \quad t' = -2.2361.$$

The *same substantive question* becomes

$$H_0 : \mu_2 - \mu_1 \geq 0, \quad H_1 : \mu_2 - \mu_1 < 0.$$

Reject if $t' < -1.734$. Again the result is significant.

Equivalent statements
“Group 1 has a higher mean than group 2” and “group 2 has a lower mean than group 1” express the same scientific alternative. Their algebraic signs differ because the subtraction order differs.

For a two-sided test, swapping the groups simply changes t to $-t$. The absolute value, p-value, and decision stay the same.

If the null difference is nonzero, reverse it too. For example, $\mu_1 - \mu_2 = 3$ becomes $\mu_2 - \mu_1 = -3$. The correct numerator is always

$$\text{observed difference} - \text{hypothesised difference},$$

with both differences written in the same order.

12.7 Illustration: Unequal Sample Sizes

Assume independent normal populations with equal variances. Samples give

$$n_1 = 8, \bar{x}_1 = 25, s_1 = 4; \quad n_2 = 10, \bar{x}_2 = 21, s_2 = \sqrt{20}.$$

Test for different population means at 5%.

$$H_0 : \mu_1 - \mu_2 = 0, \quad H_1 : \mu_1 - \mu_2 \neq 0.$$

First calculate the weighted pooled variance:

$$s_p^2 = \frac{7(16) + 9(20)}{8 + 10 - 2} = \frac{292}{16} = 18.25.$$

Then calculate the estimated standard error:

$$SE = \sqrt{18.25 \left(\frac{1}{8} + \frac{1}{10} \right)} = \sqrt{4.10625} = 2.0264.$$

Thus

$$t = \frac{25 - 21}{2.0264} = 1.9740, \quad \nu = 16.$$

Since $1.9740 < t_{0.025,16} = 2.120$, do not reject H_0 . There is insufficient evidence of a difference at the 5% two-sided level.

Flipped research question. If a higher mean in group 1 had been the pre-specified alternative, the right-sided cutoff would be 1.746, and the result would be significant. As before, the direction must not be selected after seeing the data.

Why weights matter. The two samples contribute 7 and 9 degrees of freedom. The pooled variance is not the simple average $(16 + 20)/2 = 18$ because the contributions do not have equal weights.

12.8 Worked Two-Sample Test from Raw Observations

Two independent groups use teaching methods A and B. Their scores are:

A	14	16	18	20	22
B	8	10	12	14	16

Assume independent random samples from normal populations with equal variances. Test at 5% whether the population mean scores differ.

1. Define the parameter and hypotheses

Let $\Delta = \mu_A - \mu_B$. Test $H_0 : \Delta = 0$ against $H_1 : \Delta \neq 0$. This is an independent two-sample, two-tailed pooled test.

2. Calculate the sample summaries

$$\bar{x}_A = 90/5 = 18, \quad \bar{x}_B = 60/5 = 12.$$

Both groups have deviations $-4, -2, 0, 2, 4$ from their own means, so

$$s_A^2 = s_B^2 = \frac{16 + 4 + 0 + 4 + 16}{4} = 10.$$

3. Pool the variances and find the standard error

$$s_p^2 = \frac{4(10) + 4(10)}{8} = 10, \quad \text{SE} = \sqrt{10 \left(\frac{1}{5} + \frac{1}{5} \right)} = 2.$$

4. Calculate the statistic and make the decision

$$t = \frac{(18 - 12) - 0}{2} = 3, \quad \nu = 5 + 5 - 2 = 8.$$

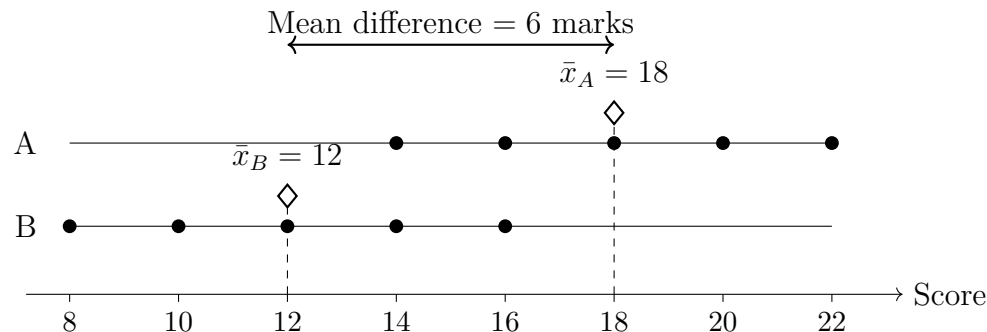
The cutoff is $t_{0.025,8} = 2.306$. Since $3 > 2.306$, reject H_0 ; the two-sided p-value is approximately 0.0171. **Conclusion.** There is evidence that the population mean scores differ. The estimated difference is 6 marks, with 95% interval

$$6 \pm 2.306(2) \approx (1.388, 10.612) \text{ marks.}$$

The stated equal-variance assumption justifies pooling; equal sample sizes alone would not justify it. The next graph shows the observations used in this calculation.

12.9 Graph: Two Independent Samples and Their Means

Each filled circle below is one observed score from the raw-data example. Open diamonds mark the sample means. The horizontal scale is shared by both groups.



What this graph shows. Group A has a higher sample mean, but the observations overlap: both groups include scores of 14 and 16. A significant mean difference does not imply that every A score exceeds every B score.

What the test adds. The mean separation of 6 marks is assessed relative to a standard error of 2 marks. Thus it is 3 standard errors above the null difference, which is sufficiently far into the tail of t_8 for the specified test.

Do not create artificial pairs

The position of a value in either list does not identify a matched person. Subtracting entries by column would produce five identical differences of 6, destroying the information about independent sampling variability. A paired test would be unjustified. The independent-sample calculation correctly uses both groups' variability.

12.10 Welch's Two-Sample t -Test: Unequal Variances

Welch's test is still a t -test for a difference of means. It estimates each sample mean's variance separately.

Welch formulas
$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}.$
Write $a = s_1^2/n_1$ and $b = s_2^2/n_2$. Then
$\nu \approx \frac{(a + b)^2}{a^2/(n_1 - 1) + b^2/(n_2 - 1)}.$
This is the Welch–Satterthwaite approximation.

The hypotheses and tail rules have the same form as in the pooled test. What changes is the estimated standard error and the degrees of freedom.

Feature	Pooled t	Welch's t
Equal population variances	Assumed	Not required
Variance estimate	Combined s_p^2	Separate s_1^2, s_2^2
Degrees of freedom	$n_1 + n_2 - 2$	Approximation above
Independent samples	Required	Required

Software can use non-integer degrees of freedom directly. With a printed table, using a smaller available df gives a conservative cutoff; state the approximation.

For an elementary examination problem explicitly stating equal variances, use the pooled formula. If that assumption is absent and Welch is permitted, it is usually a sensible independent-sample choice. Welch and pooled tests answer the same population-mean question under different variance assumptions.

12.11 Illustration: Welch Calculation

Two independent samples give

$$n_1 = 10, \bar{x}_1 = 56, s_1 = 8; \quad n_2 = 12, \bar{x}_2 = 50, s_2 = 4.$$

Assume suitable normal-population conditions, without assuming equal variances. Test for a difference at 5%.

$$H_0 : \mu_1 - \mu_2 = 0, \quad H_1 : \mu_1 - \mu_2 \neq 0.$$

Calculate

$$a = \frac{64}{10} = 6.4, \quad b = \frac{16}{12} = 1.3333.$$

The standard error and statistic are

$$SE = \sqrt{6.4 + 1.3333} = 2.7809, \quad t = \frac{6}{2.7809} = 2.1576.$$

The degrees of freedom are approximately

$$\nu = \frac{(6.4 + 1.3333)^2}{6.4^2/9 + 1.3333^2/11} = 12.6900.$$

Using this df, the two-sided critical value is approximately 2.1657. Since $2.1576 < 2.1657$, do not reject H_0 at 5%.

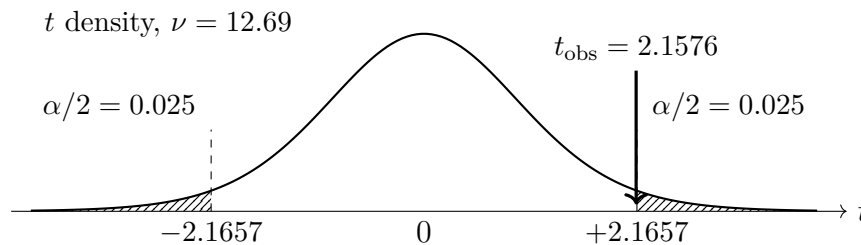
The approximate two-sided p-value is 0.05075. This is a close result, which is why retaining sufficient precision matters.

Using the conservative printed-table row $\nu = 12$ gives cutoff 2.179 and the same decision. Do not replace the Welch df by $n_1 + n_2 - 2 = 20$; that belongs to the pooled model.

Conclusion. The evidence is just short of the chosen threshold. It is more informative to report the estimated difference of 6 units and the p-value than to describe the result simply as “no difference”.

12.12 Graph: Welch's Test and the Interval for the Difference

The Welch example gives a mean difference of 6, standard error 2.7809, and approximate degrees of freedom 12.6900.

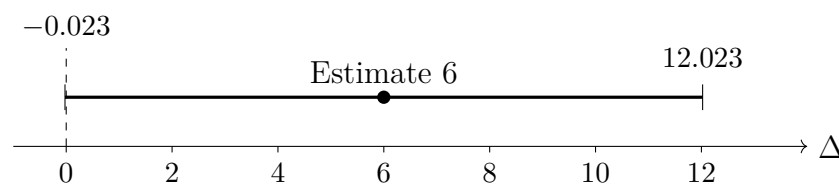


The observed statistic is just inside the non-rejection region. The observed value and cutoff are very close, so use their numbers to resolve the comparison. The density is plotted on a numerical t scale. Each rejection tail continues beyond the displayed range.

The 95% Welch interval is

$$6 \pm 2.1657(2.7809) \approx (-0.023, 12.023).$$

The endpoints use unrounded intermediate values. The interval includes 0, agreeing with the two-sided p -value $p \approx 0.05075$.



Numerically scaled 95% interval for $\mu_1 - \mu_2$; the left endpoint is just below zero.

Interpretation. The interval spans a small negative difference and substantial positive differences. Non-rejection does not establish equal means. The estimate, interval, and p -value describe the result more fully than the decision alone.

12.13 Two-Sample Test with a Nonzero Benchmark

A supplier wants evidence that product A lasts, on average, more than 2 hours longer than product B. Independent samples give

$$n_A = n_B = 10, \quad \bar{x}_A = 52, \quad \bar{x}_B = 48, \quad s_A = s_B = 4.$$

Assume normal populations with equal variances. Define $\Delta = \mu_A - \mu_B$.

$$H_0 : \Delta \leq 2, \quad H_1 : \Delta > 2, \quad \alpha = 0.05.$$

The pooled variance and standard error are $s_p^2 = 16$ and $SE = 1.7889$. The numerator must subtract the claimed improvement:

$$t = \frac{(52 - 48) - 2}{1.7889} = 1.1180, \quad \nu = 18.$$

Reject only if $t > t_{0.05,18} = 1.734$. Since $1.1180 < 1.734$, do not reject H_0 . There is insufficient evidence that A's population mean life exceeds B's by more than 2 hours.

Different questions have different null boundaries
--

The same samples support a nonzero mean difference in the earlier two-sided test. That finding does not automatically establish a difference greater than 2 hours. Subtracting Δ_0 makes the calculation answer the actual question.

If the subtraction order is reversed, define $\Delta' = \mu_B - \mu_A$. The same claim becomes $H_1 : \Delta' < -2$, with null $H_0 : \Delta' \geq -2$. The statistic is -1.1180 and the cutoff is -1.734 , giving the same conclusion.

12.14 Two-Sample t -Test: Complete Working Procedure

1. **Check the design.** Use independent samples for unrelated groups; use the paired test for repeated measurements or genuine matched pairs.
2. **Define Δ .** State which population mean is subtracted from which.
3. **Set the hypotheses and level.** Choose H_1 from the question, include equality in H_0 , and record Δ_0 and α .
4. **Select the variance model.** Pool only with a justified common population variance; otherwise use Welch under suitable sampling conditions.
5. **Calculate both sample summaries.** Record n_i , $\bar{x}_i$, and s_i^2 for each group; do not confuse SD and variance.
6. **Find the matching SE and df.** Keep pooled SE with pooled df, or Welch SE with Welch df.
7. **Standardise the difference.**

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{\text{SE}}.$$

8. **Apply the correct tail rule.** Use the signed statistic for a one-sided test and $|t|$ for the symmetric two-sided test.
9. **Report in context.** State the decision, estimated difference, and evidence about the population means. Include a matching interval when useful.

Confidence interval for two independent population means

For a two-sided level- α test, use

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, \nu} \text{SE}.$$

The standard error and df come from the selected pooled or Welch method. Compare the interval with Δ_0 , which need not be zero.

The next section develops the confidence-interval connection for all three designs. Later practice questions let you apply the same sequence independently.

13 Confidence Intervals and t -Tests

13.1 Two Ways to Describe the Same Evidence

A two-sided $100(1 - \alpha)\%$ confidence interval for a normal-population mean with unknown standard deviation is

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}.$$

For the corresponding two-sided level- α test:

- If μ_0 lies strictly outside the interval, reject $H_0 : \mu = \mu_0$.
- If μ_0 lies inside the interval, do not reject H_0 .

Exact equality at an endpoint corresponds to the critical boundary; use the stated boundary convention and full precision in that rare case.

Why the connection works

The two-sided non-rejection inequality is

$$\left| \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \right| \leq c.$$

Rearranging gives

$$\bar{x} - c \frac{s}{\sqrt{n}} \leq \mu_0 \leq \bar{x} + c \frac{s}{\sqrt{n}}.$$

Those are precisely the confidence-interval endpoints.

For paired data, replace $\bar{x}$ and s by $\bar{d}$ and s_d . For independent samples, use the estimated mean difference and its matching standard error and df.

Interpretation. A 95% confidence procedure produces intervals that cover the fixed population parameter in 95% of repeated suitable samples. After this interval is observed, the usual frequentist interpretation is not that the fixed parameter itself now has a 95% probability of lying inside it.

13.2 Illustration: Mean and Difference Intervals

For the bottle example,

$$\bar{x} = 496, \quad \text{SE} = 1.5811, \quad \nu = 9.$$

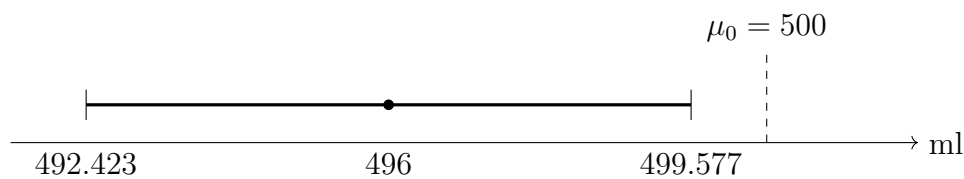
The 95% interval is

$$496 \pm 2.2622(1.5811) = 496 \pm 3.5768,$$

or approximately

$$(492.423, 499.577) \text{ ml.}$$

The null value 500 is outside the interval, agreeing with rejection in the two-sided 5% test.



Schematic interval diagram; endpoint values are labelled.

For the pooled example with difference 4, $\text{SE} = 1.7889$, and $\nu = 18$,

$$4 \pm 2.1009(1.7889) \approx 4 \pm 3.7582.$$

The interval for $\mu_1 - \mu_2$ is approximately

$$(0.242, 7.758).$$

It excludes 0, again agreeing with the two-sided decision.

Use the interval for the *difference*. Comparing whether two separate group confidence intervals overlap is not the same decision rule as the two-sample t -test.

14 Understanding the Result

14.1 Statistical Significance and Practical Importance

A statistically significant result means that the observation passes the chosen test rule. It does not by itself tell us whether the difference is large enough to matter in practice.

Suppose a large study detects an average time reduction of 0.02 seconds. The reduction may be statistically convincing but operationally unimportant. Conversely, a small study may estimate a potentially useful reduction with so much uncertainty that it is not statistically significant.

Question	Quantity to examine	Purpose
Is there evidence against the null?	t , df, and p-value	Statistical evidence
How large is the observed change?	Mean difference in original units	Size of the estimate
How uncertain is the estimate?	Standard error or confidence interval	Precision
Does the change matter?	A context-specific benchmark	Practical importance

A useful reporting sentence

“The estimated mean increase was 4 units. The pooled test gave $t = 2.236$ with 18 degrees of freedom, significant at 5%. The 95% confidence interval for the population mean difference was approximately 0.242 to 7.758 units.”

This sentence gives the result, its scale, and its uncertainty. A bare statement such as “ H_0 rejected” gives the reader much less information.

Also remember that significance does not establish causation. Random sampling supports population generalisation; appropriate random assignment and experimental control support causal interpretation. These are different design questions.

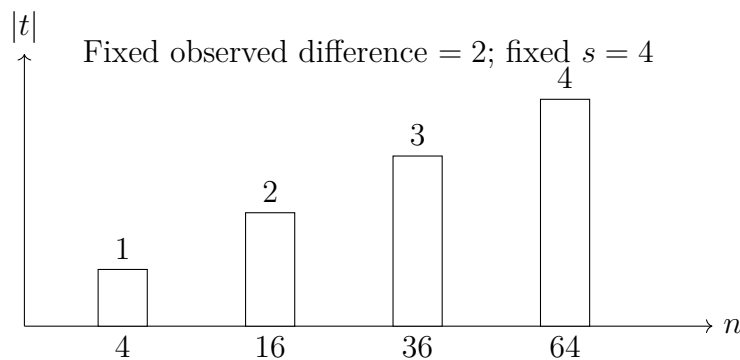
14.2 What Makes the t Statistic Larger?

For one sample,

$$|t| = \frac{|\bar{x} - \mu_0|\sqrt{n}}{s}.$$

Holding the other quantities fixed:

- a larger observed difference increases $|t|$;
- a larger sample size increases $|t|$;
- a larger sample standard deviation decreases $|t|$.



This chart is a controlled numerical comparison, not a prediction that every larger real sample will produce a larger statistic. In actual new samples, $\bar{x}$ and s also change.

The sample size additionally changes df and therefore the critical value. Increasing n usually improves the ability to detect a specified true departure under a fixed design, but sample quality and independence remain essential.

Do not keep collecting observations and repeatedly testing until a significant result appears under an ordinary fixed-sample rule. Such a procedure is different from the test whose significance level was specified at the start.

15 Mixed Worked Illustrations

15.1 Illustration: Select the Design before the Formula

Three questions each mention a mean score of 60.

Case A – Fixed benchmark

A random sample of 12 current students is compared with a specified population benchmark of 60.

Use a one-sample test, with numerator $\bar{x} - 60$ and df 11.

Case B – Same students measured twice

The same 12 students are tested before and after a course. Their before mean is 60.

Use a paired test. Calculate each student's change, then use $\bar{d}$, s_d , and df 11. The before mean alone is not enough to obtain s_d .

Case C – A different comparison group

Twelve current students are compared with ten different students whose sample mean is 60.

Use an independent-sample test. Include uncertainty from both groups. If equal population variances are assumed, use pooled df $12 + 10 - 2 = 20$; otherwise use Welch.

The appearance of the number 60 does not determine the method. Its role does: fixed reference, mean of paired baseline measurements, or mean of an independent sample.

Diagnostic question. Ask “Where did the comparison number come from?” before treating it as a known reference. This prevents the common error of using a one-sample denominator when the comparison quantity was estimated from data.

15.2 Illustration: A Nonzero Mean-Change Benchmark

A training programme is considered worthwhile only if it raises the mean score by more than 2 marks. Use the eight paired coaching observations, for which

$$\bar{d} = 3.5, \quad s_d = 1.1952, \quad n = 8.$$

Here $d = \text{after} - \text{before}$. The appropriate hypotheses are

$$H_0 : \mu_d \leq 2, \quad H_1 : \mu_d > 2.$$

The null-boundary difference is $\Delta_0 = 2$, not 0.

$$t = \frac{3.5 - 2}{1.1952/\sqrt{8}} = \frac{1.5}{0.4226} = 3.5496.$$

With $\nu = 7$, reject if $t > 1.895$. Therefore, reject H_0 at 5%. There is sufficient evidence that the population mean increase exceeds 2 marks.

Flipped benchmark. Suppose the requirement were an increase of more than 3 marks instead. Then

$$t = \frac{3.5 - 3}{0.4226} = 1.1832 < 1.895.$$

The evidence would be insufficient for an increase exceeding 3 marks.

What changed?
The same sample can support “mean increase above 0” and “mean increase above 2” while not supporting “mean increase above 3”. These are different null boundaries, so the numerator must change accordingly.

Always read the full benchmark in the question. Automatically subtracting zero in every paired or two-sample test can answer the wrong question even when all other arithmetic is correct.

15.3 Illustration: Check Whether the Calculation Is Justified

Scenario 1: repeated readings of one object. A student measures the same single rod 20 times and treats the readings as 20 independent rods from a factory.

The readings describe repeated measurement of one object. They do not supply 20 independent production units. A t -test cannot justify generalising to all factory rods using that claimed sample size.

Scenario 2: arbitrary pairing. Ten observations from machine A and ten from machine B are sorted separately and paired by rank, although the units were sampled independently.

This creates artificial pairs. Use the independent-sample design; do not choose a paired method merely because it gives a smaller standard error.

Scenario 3: standard deviations alone for a paired test. The before and after means and SDs are known, but the individual differences or their variance are not supplied.

The paired standard error cannot generally be recovered from those summaries alone. It also depends on the relationship between the two measurements:

$$\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2 \text{Cov}(A, B).$$

Main lesson
The formula is the last part of the design decision. Correctly identifying the independent observations and the measured parameter comes first.

When needed information is missing, identify it explicitly. Do not make up pair differences or silently assume that paired measurements are independent.

16 Common Errors

16.1 Errors in Hypotheses and Decisions

Error 1 – Writing hypotheses about the sample mean. Incorrect: $H_0 : \bar{x} = 50$. The sample mean is already observed. Correct: $H_0 : \mu = 50$, a claim about the unknown population mean.

Error 2 – Choosing the tail from the observed result. A low sample mean does not turn a pre-specified two-sided question into a left-sided test. Use the original research question.

Error 3 – Placing equality in the alternative. For the ordinary tests here, equality is included in H_0 . Use $>$, $<$, or $\neq$ for H_1 .

Error 4 – Taking an absolute value in every test. Absolute value is used in the usual two-sided comparison. Direction matters for one-sided tests.

Error 5 – Saying that non-rejection proves the null. Insufficient evidence of a difference does not establish equality. The study may have limited precision or power.

Error 6 – Treating the p-value as the probability of the null. The p-value is calculated under the null model; it is not a probability assigned to the hypothesis being true.

Quick correction drill

Statement: “Since $p = 0.08$, there is an 8% chance that the null is true.”

Correction: “Under the null model, a result at least as extreme as the observed one has probability 0.08. At $\alpha = 0.05$, we do not reject H_0 .”

These wording errors are important because they change the meaning of an otherwise correct numerical solution.

16.2 Errors in Calculation and Test Selection

Mistake	Correction
Using s as the standard error	Use $s/\sqrt{n}$ for one mean.
Using variance directly as SD	Take the square root first.
Using divisor n in sample variance	Use $n - 1$ for the usual s^2 .
Using n as one-sample df	Use $n - 1$.
Using $2n - 2$ for n pairs	Use $n - 1$ for paired differences.
Using $q = 0.05$ for a two-sided 5% test	Use $q = 0.025$.
Using a positive cutoff for a left tail	Reject below the negative cutoff.
Averaging two SDs to pool spread	Pool df-weighted variances.
Using pooled df with Welch's SE	Use the matching Welch df.
Ignoring the sign convention for differences	Define subtraction before hypotheses.
Rounding each intermediate number heavily	Keep precision until reporting.

A three-part final check

Sign: does the statistic point in the expected direction?

Scale: is the denominator a standard error in the same units as the numerator?

Rule: do the df, tail, and significance level match the chosen test?

If your conclusion changes when you merely swap group labels and consistently reverse the alternative, an error has occurred. Equivalent formulations must produce the same scientific decision.

17 Practice Questions

17.1 Foundations

Try these questions without looking at the answers. For numerical questions, assume independent observations and suitable normal-population conditions unless the question asks you to examine those conditions. Use $\alpha = 0.05$ unless another level is specified.

Q1. Define population, sample, parameter, and statistic. Give an example of each using the filling-volume problem.

Q2. Explain why a sample mean different from a claimed mean does not, by itself, prove that the population claim is false.

Q3. Distinguish H_0 from H_1 . Explain why hypotheses for a population mean should use μ rather than $\bar{x}$.

Q4. A sample has $n = 16$ and $s = 8$. Find the estimated standard error of the mean and the one-sample degrees of freedom. Explain the difference between the numbers s and SE.

Q5. For a two-sided 5% test with $\nu = 9$, identify the upper-tail table column and both critical values. Repeat for a left-sided 5% test.

Q6. A test has $p = 0.12$. State the decision at 5%. Explain why this does not establish that H_0 is true.

Writing practice

Answer conceptual questions in complete sentences. A formula alone may not explain the difference between a sample observation and a population conclusion.

Self-check before moving on: can you explain what the denominator of the t statistic measures without simply repeating the formula? It estimates the sampling uncertainty of the mean or mean difference being tested.

17.2 Identifying and Flipping Hypotheses

For Q7–Q12, define the parameter, state both hypotheses, identify the tail, and state the null-boundary value used in the calculation.

Q7. A machine is intended to produce rods with mean length 20 cm. Test whether the population mean length has changed.

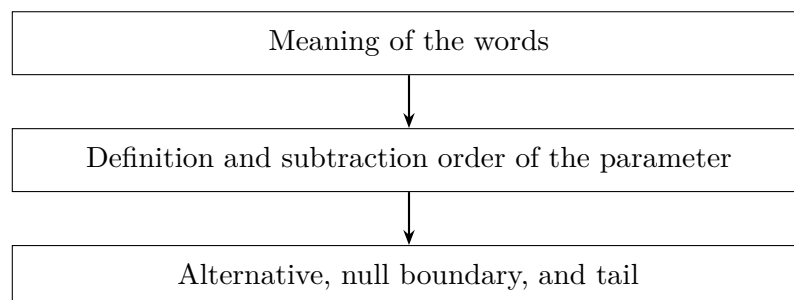
Q8. A battery manufacturer claims a population mean life of at least 100 hours. A customer suspects a lower mean.

Q9. A service claims a population mean waiting time of at most 15 minutes. A customer suspects that the mean waiting time is longer.

Q10. The same people complete a task before and after training. Define $d = \text{after time} - \text{before time}$. State hypotheses for evidence that the training reduces mean completion time.

Q11. Repeat Q10 after redefining the difference as $d' = \text{before time} - \text{after time}$. Explain what should happen to the test statistic and final substantive conclusion.

Q12. Two independent teaching methods are compared. Let μ_A and μ_B denote population mean scores. The aim is to establish that method A exceeds method B by more than 3 marks.



17.3 One-Sample Calculations

For each question, show hypotheses, the standard error, t , df, the critical comparison, and a conclusion in the context of the question.

Q13. A sample of 9 items has mean mass 104 g and sample standard deviation 6 g. Test whether the population mean differs from 100 g.

Q14. Use the same data as Q13, but suppose that evidence of a mean above 100 g was the pre-specified research question. Compare the decision with Q13.

Q15. A process has a benchmark mean completion time of 50 minutes. A sample of 16 observations gives $\bar{x} = 48$ and $s = 4$. Test for a reduction in the population mean time.

Q16. A sample of 25 observations has $\bar{x} = 102$ and $s = 10$. Test $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$.

Q17. The observations are 9, 10, 11, 12, 13. Test whether the population mean differs from 10. Calculate the sample variance explicitly.

Q18. A sample is summarised by $n = 10$, $\sum x_i = 210$, and $\sum x_i^2 = 4450$. Test for a population mean greater than 20.

Show the working

For Q17 and Q18, finding the correct sample variance is part of the solution. Do not skip directly to a final statistic. For Q13 and Q14, explain why a change in the alternative changes the critical value even though the data are identical.

17.4 Paired Calculations

Q19. Twelve students are tested before and after a course. How many differences are used in a paired test? What are the degrees of freedom? Why is this not automatically an independent two-sample problem?

Q20. Five independent pairs give after-minus-before score differences 1, 2, 3, 2, 2. Test for a positive population mean change.

Q21. Reverse the subtraction in Q20. Give the new mean difference, sample SD, test statistic, alternative, and decision.

Q22. Five pairs give differences $-1, 0, 1, 0, 0$. Test for any population mean difference from zero.

Q23. Using the differences in Q20, test whether the population mean increase exceeds 1 mark rather than merely exceeding zero.

Q24. Four people have before times 20, 25, 30, 35 minutes and corresponding after times 18, 24, 27, 34 minutes. Use before minus after and test for a reduction in population mean time.

A table layout for paired working

Pair	Before	After	d_i	d_i^2
1	measurement	measurement	difference	square
2	measurement	measurement	difference	square
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Label the subtraction order before filling the difference column.

Use the number of complete pairs as n . In Q24, there are four differences, even though eight individual time measurements are listed.

17.5 Two-Sample t -Test Calculations

For Q25–Q28 assume independent normal populations with equal variances.

Q25. Group 1 has $n_1 = 10$, $\bar{x}_1 = 52$, $s_1 = 4$. Group 2 has $n_2 = 10$, $\bar{x}_2 = 48$, $s_2 = 4$. Test whether the population means differ.

Q26. Group 1 has $n_1 = 8$, $\bar{x}_1 = 30$, $s_1 = 4$. Group 2 has $n_2 = 8$, $\bar{x}_2 = 26$, $s_2 = \sqrt{24}$. Test whether the population means differ.

Q27. For the data in Q26, instead use the pre-specified alternative that group 1 has the higher population mean.

Q28. Group 1 has $n_1 = 8$, $\bar{x}_1 = 25$, $s_1 = 4$. Group 2 has $n_2 = 10$, $\bar{x}_2 = 21$, $s_2 = \sqrt{20}$. Carry out a two-sided test. Explain why the pooled variance uses unequal weights.

Q29. Use Welch's test for independent samples with $n_1 = n_2 = 10$, $\bar{x}_1 = 15$, $\bar{x}_2 = 12$, $s_1 = 3$, and $s_2 = 4$. Test for a difference. Calculate the approximate df; use the conservative table row $\nu = 16$ if necessary.

Q30. In Q25, reverse the group labels. What happens to the pooled variance, standard error, signed statistic, and two-sided conclusion?

Reminder. Use s_1^2 and s_2^2 in the pooled variance formula. If $s_2 = \sqrt{24}$, then $s_2^2 = 24$. Carry the radical exactly until numerical evaluation is needed.

17.6 Interpretation and Error Detection

Q31. A two-sided test has $p = 0.032$. State the decision at 10%, 5%, and 1%. Does the p-value change when the chosen significance level changes?

Q32. For $t = -2$ and $\nu = 9$, the lower-tail probability is approximately 0.0383. Find the p-value for a left-sided, right-sided, and two-sided alternative.

Q33. A 95% t confidence interval for a population mean is (492.423, 499.577) ml. What is the corresponding two-sided 5% test decision for $H_0 : \mu = 500$? What about $H_0 : \mu = 496$?

Q34. Critique the statement: “The p-value is 0.40, so the two population means have been proved equal.”

Q35. For $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$, describe Type I and Type II errors in words. What does reducing α generally do to power if everything else in the study design is fixed?

Q36. A student observes five measurements, one of which is an extreme outlier. After seeing that the sample mean is above the benchmark, the student selects a right-tailed test and declares the method valid because “t-tests are for small samples”. Identify two problems with this reasoning.

Before checking the answers
For every numerical answer, check the sign, the denominator, the degrees of freedom, and the tail. For every verbal answer, check whether you are making a claim about the sample, the population, or the probability model.

17.7 Additional Two-Sample Practice

Q37. Two independent samples give A: 14, 16, 18, 20, 22 and B: 8, 10, 12, 14, 16. Assume normal populations with equal variances. Carry out a two-sided pooled test at 5% and find a 95% interval for $\mu_A - \mu_B$. Explain why overlap between individual scores does not decide the test.

Q38. Use the summaries in Q25 to test the pre-specified claim that $\mu_1 - \mu_2 > 2$ at 5%. Explain why the numerator is not $52 - 48$.

Q39. For independent samples with $n_1 = 10$, $\bar{x}_1 = 56$, $s_1 = 8$, $n_2 = 12$, $\bar{x}_2 = 50$, $s_2 = 4$, use Welch's test at the two-sided 5% level. Use $\nu \approx 12.6900$ and $t_{0.025, \nu} \approx 2.1657$ to compute a 95% interval. Explain what inclusion of zero means.

Q40. Two unrelated groups each have ten observations. A student pairs the first observation in one group with the first in the other, and so on. Why is this invalid? If equal population variances are not justified, which independent two-sample method can be used? Does “two-sample” determine the tail?

Sketch before concluding

For Q37 and Q39, draw an axis for Δ , mark zero, and plot the confidence interval. For Q38, sketch a right-tailed t_{18} curve and label the observed statistic and the positive critical value. Your graph and conclusion should agree.

18 Answers to Practice Questions

18.1 Answers Q1–Q6: Foundations

Q1. The population is the full collection of bottles produced under the process of interest. The sample is the selected set actually measured. The population mean volume μ is a parameter. The calculated sample mean $\bar{x}$ is a statistic.

Q2. Different random samples usually have different sample means even when the population mean is fixed. A departure must be judged relative to its estimated standard error and the reference distribution. Mere inequality $\bar{x} \neq \mu_0$ is not enough for rejection.

Q3. H_0 is the statement assessed against the evidence; H_1 specifies the departure being investigated. Hypotheses concern the unknown population parameter μ . The observed $\bar{x}$ is information used to test them.

Q4. The estimated standard error is

$$SE = \frac{8}{\sqrt{16}} = 2, \quad \nu = 16 - 1 = 15.$$

The SD 8 describes individual variation. The SE 2 estimates the sampling uncertainty of the mean.

Q5. For the two-sided test, use $q = 0.025$, giving cutoffs -2.262 and $+2.262$. For the left-sided test, use $q = 0.05$ and reject below -1.833 .

Q6. Because $0.12 > 0.05$, do not reject H_0 . The evidence is insufficient at the chosen level. A false null can also fail to be rejected, so this decision is not proof that the null is true.

Core idea
A hypothesis test compares a sample departure with its uncertainty. Both the numerical comparison and the interpretation must refer to the specified population question.

18.2 Answers Q7–Q12: Hypotheses

Q7. Let μ be the population mean rod length:

$$H_0 : \mu = 20, \quad H_1 : \mu \neq 20.$$

Two-tailed; null reference $\mu_0 = 20$ cm.

Q8. Let μ be the population mean battery life:

$$H_0 : \mu \geq 100, \quad H_1 : \mu < 100.$$

Left-tailed; calculate at the boundary 100 hours.

Q9. Let μ be the population mean waiting time:

$$H_0 : \mu \leq 15, \quad H_1 : \mu > 15.$$

Right-tailed; boundary 15 minutes.

Q10. For after minus before, reduced time means negative differences:

$$H_0 : \mu_d \geq 0, \quad H_1 : \mu_d < 0.$$

Left-tailed; boundary 0.

Q11. For before minus after, reduced time means positive differences:

$$H_0 : \mu_{d'} \leq 0, \quad H_1 : \mu_{d'} > 0.$$

Right-tailed; boundary 0. The statistic changes sign, the SD is unchanged, and the substantive conclusion is unchanged.

Q12. Define $\Delta = \mu_A - \mu_B$. Then

$$H_0 : \Delta \leq 3, \quad H_1 : \Delta > 3.$$

Right-tailed; use $\Delta_0 = 3$ in the numerator.

18.3 Answers Q13–Q15: One Sample

Q13. Use $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$.

$$SE = \frac{6}{\sqrt{9}} = 2, \quad t = \frac{104 - 100}{2} = 2, \quad \nu = 8.$$

The two-sided cutoff is 2.306. Since $2 < 2.306$, do not reject H_0 . There is insufficient evidence that the population mean mass differs from 100 g. The two-sided p-value is approximately 0.0805.

Q14. Use $H_0 : \mu \leq 100$ against $H_1 : \mu > 100$. The same data give $t = 2$ and $\nu = 8$, but the right-sided cutoff is 1.860.

$$2 > 1.860.$$

Reject H_0 . There is sufficient evidence of a population mean mass above 100 g. The right-sided p-value is approximately 0.0403. The conclusion differs from Q13 because the pre-specified alternatives differ.

Q15. Let μ be population mean completion time. Use $H_0 : \mu \geq 50$ against $H_1 : \mu < 50$.

$$SE = \frac{4}{\sqrt{16}} = 1, \quad t = \frac{48 - 50}{1} = -2, \quad \nu = 15.$$

Reject below -1.753 . Since $-2 < -1.753$, reject H_0 . There is sufficient evidence that the population mean time is below 50 minutes. The left-sided p-value is approximately 0.0320.

Comparison

The sign tells us the direction of the departure. The critical rule tells us whether that departure is sufficiently large for the question being tested.

18.4 Answers Q16–Q18: One-Sample Working

Q16. Use $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$.

$$SE = \frac{10}{5} = 2, \quad t = \frac{102 - 100}{2} = 1, \quad \nu = 24.$$

Since $1 < 2.064$, do not reject H_0 . There is insufficient evidence of a population mean different from 100; the two-sided p -value is about 0.3273.

Q17. The mean is $\bar{x} = 55/5 = 11$. Deviations from 11 are $-2, -1, 0, 1, 2$, with squared deviations summing to 10.

$$s^2 = \frac{10}{4} = 2.5, \quad SE = \sqrt{2.5/5} = 0.7071.$$

For $H_0 : \mu = 10$ against $H_1 : \mu \neq 10$,

$$t = \frac{11 - 10}{0.7071} = 1.4142, \quad \nu = 4.$$

Since $1.4142 < 2.776$, do not reject H_0 . There is insufficient evidence of a population mean different from 10.

Q18. The summaries give

$$\bar{x} = \frac{210}{10} = 21, \quad s^2 = \frac{4450 - 210^2/10}{9} = \frac{40}{9}.$$

Hence $SE = \sqrt{(40/9)/10} = 2/3$. With $H_0 : \mu \leq 20$ and $H_1 : \mu > 20$,

$$t = \frac{21 - 20}{2/3} = 1.5, \quad \nu = 9.$$

Since $1.5 < 1.833$, do not reject H_0 . The evidence is insufficient for a population mean above 20.

18.5 Answers Q19–Q22: Paired Tests

Q19. There are 12 paired differences and 11 degrees of freedom. Measurements on the same student are related, so they should not be treated as two unrelated groups. The test is performed on the within-student changes.

Q20. Define $d = \text{after} - \text{before}$ and test $H_0 : \mu_d \leq 0$ against $H_1 : \mu_d > 0$.

$$\sum d_i = 10, \quad \sum d_i^2 = 22, \quad \bar{d} = 2, \quad s_d^2 = \frac{22 - 10^2/5}{4} = 0.5.$$

$$\text{SE} = \sqrt{0.5/5} = 0.3162, \quad t = 6.3246, \quad \nu = 4.$$

Since $6.3246 > 2.132$, reject H_0 . There is sufficient evidence of a positive population mean change.

Q21. The new differences are $-1, -2, -3, -2, -2$.

$$\bar{d}' = -2, \quad s_{d'} = \sqrt{0.5}, \quad t' = -6.3246.$$

Use $H_0 : \mu_{d'} \geq 0$ against $H_1 : \mu_{d'} < 0$. Since $-6.3246 < -2.132$, reject. The evidence still supports increased scores.

Q22. For $H_0 : \mu_d = 0$ against $H_1 : \mu_d \neq 0$,

$$\bar{d} = 0, \quad s_d^2 = 2/4 = 0.5, \quad t = 0, \quad \nu = 4.$$

Since $0 < 2.776$, do not reject H_0 . There is no evidence in these data of a nonzero population mean difference; the two-sided p -value is 1. This does not prove that every pair has zero change.

18.6 Answers Q23–Q26: Differences of Means

Q23. Use $H_0 : \mu_d \leq 1$ against $H_1 : \mu_d > 1$. From Q20,

$$t = \frac{2 - 1}{0.3162} = 3.1623, \quad \nu = 4.$$

Since $3.1623 > 2.132$, reject H_0 . The evidence supports a population mean increase exceeding 1 mark.

Q24. Before-minus-after differences are 2, 1, 3, 1. Thus

$$\bar{d} = 7/4 = 1.75, \quad s_d^2 = \frac{15 - 7^2/4}{3} = \frac{11}{12}.$$

With $H_0 : \mu_d \leq 0$ against $H_1 : \mu_d > 0$,

$$SE = \sqrt{11/48} = 0.4787, \quad t = 3.6556, \quad \nu = 3.$$

Since $3.6556 > 2.353$, reject. There is sufficient evidence of reduced population mean completion time.

Q25. Test $H_0 : \mu_1 - \mu_2 = 0$ against $H_1 : \mu_1 - \mu_2 \neq 0$.

$$s_p^2 = 16, \quad SE = \sqrt{3.2} = 1.7889, \quad t = 2.2361, \quad \nu = 18.$$

Since $2.2361 > 2.101$, reject. There is evidence that the population means differ.

Q26. Use the same two-sided hypothesis form.

$$s_p^2 = \frac{7(16) + 7(24)}{14} = 20, \quad SE = \sqrt{20(1/8 + 1/8)} = \sqrt{5}.$$

Thus $t = 4/\sqrt{5} = 1.7889$ and $\nu = 14$. Since $1.7889 < 2.145$, do not reject. There is insufficient evidence of a population mean difference at 5%.

18.7 Answers Q27–Q30: Independent Samples

Q27. Use $H_0 : \mu_1 - \mu_2 \leq 0$ against $H_1 : \mu_1 - \mu_2 > 0$. The statistic is still 1.7889 with 14 df. The right-tailed cutoff is 1.761. Since $1.7889 > 1.761$, reject. There is evidence that group 1 has a higher population mean under the pre-specified one-sided question.

Q28. For $H_0 : \mu_1 - \mu_2 = 0$ against $H_1 : \mu_1 - \mu_2 \neq 0$,

$$s_p^2 = \frac{7(16) + 9(20)}{16} = 18.25, \quad \text{SE} = 2.0264, \quad t = 1.9740, \quad \nu = 16.$$

Since $1.9740 < 2.120$, do not reject. The evidence is insufficient for different population means. The variance weights 7 and 9 are the sample-specific df.

Q29. For the two-sided equal-mean null, Welch's formula gives

$$\text{SE} = \sqrt{9/10 + 16/10} = \sqrt{2.5}, \quad t = 1.8974.$$

$$\nu \approx \frac{(0.9 + 1.6)^2}{0.9^2/9 + 1.6^2/9} = 16.6914.$$

Using conservative df 16, the critical value is 2.120. Do not reject, since $1.8974 < 2.120$. The non-integer-df calculation also gives non-rejection. There is insufficient evidence that the population means differ.

Q30. The pooled variance remains 16 and the standard error remains 1.7889. The statistic changes from +2.2361 to -2.2361 . The two-sided p-value and decision remain unchanged, because $|t|$ is unchanged. The estimated difference now describes group 2 minus group 1.

18.8 Answers Q31–Q36: Interpretation

Q31. Reject at 10% and 5% because 0.032 is below both thresholds. Do not reject at 1%. The p-value stays 0.032 when only α changes.

Q32. Left-sided: $p \approx 0.0383$. Right-sided: $p \approx 1 - 0.0383 = 0.9617$. Two-sided: $p \approx 2(0.0383) = 0.0766$. At 5%, only the left-sided question leads to rejection.

Q33. Reject $H_0 : \mu = 500$ because 500 lies outside the interval. Do not reject $H_0 : \mu = 496$ because 496 lies inside it. These statements use the matching two-sided 5% tests and the same assumptions as the interval.

Q34. A large p-value indicates insufficient evidence against the specified null model. It does not prove equality. The observed mean difference and an interval for it help show which population differences remain plausible.

Q35. Type I error: conclude that the mean differs from 50 when it is 50. Type II error: fail to reject when the true mean differs from 50. Reducing α makes rejection more demanding and generally reduces power against a given alternative if the rest of the design is fixed.

Q36. First, the tail was chosen after seeing the sample direction, which invalidates the ordinary pre-specified one-sided error control. Second, a tiny sample with an extreme outlier does not automatically satisfy the conditions for reliable t inference. “Small sample” is not a substitute for independence and an appropriate distributional model.

What a good answer preserves

It preserves the direction of the claim, the sampling design, the distinction between evidence and proof, and the uncertainty in the estimate.

18.9 Answers Q37–Q40: Complete Two-Sample Inference

Q37. For $H_0 : \mu_A - \mu_B = 0$ against $H_1 : \mu_A - \mu_B \neq 0$,

$$\bar{x}_A = 18, \quad \bar{x}_B = 12, \quad s_A^2 = s_B^2 = 10, \quad s_p^2 = 10.$$

Thus $SE = 2$, $t = 3$, and $\nu = 8$. Since $3 > 2.306$, reject H_0 ; $p \approx 0.0171$. The interval is $6 \pm 2.306(2) = (1.388, 10.612)$ marks. There is evidence of different population means. Individual observations may overlap even when the mean difference is statistically significant.

Q38. Define $\Delta = \mu_1 - \mu_2$. Use $H_0 : \Delta \leq 2$ against $H_1 : \Delta > 2$. With $s_p^2 = 16$ and $SE = 1.7889$,

$$t = \frac{4 - 2}{1.7889} = 1.1180, \quad \nu = 18.$$

Since $1.1180 < 1.734$, do not reject. The evidence is insufficient for a population difference greater than 2. Using 4 alone in the numerator tests the boundary $\Delta_0 = 0$, which is a different question.

Q39. For the equal-mean null and two-sided alternative,

$$SE = \sqrt{64/10 + 16/12} = 2.7809, \quad t = 2.1576.$$

Since $2.1576 < 2.1657$, do not reject at 5%; $p \approx 0.05075$. The interval is $6 \pm 2.1657(2.7809) \approx (-0.023, 12.023)$. Zero is included, so equal means are not rejected by the matching test. This does not prove equality or rule out substantial positive differences.

Q40. Pairing must reflect the study design, not arbitrary positions in a list or equal sample sizes. Artificial pairing uses a dependence structure that the design does not justify. Welch's independent two-sample test avoids assuming equal population variances, subject to the other conditions. The alternative determines the tail: two-sample tests can be left-, right-, or two-tailed.

19 Formula Sheet

19.1 Foundations and One-Sample Test

Sample summaries

$$\bar{x} = \frac{\sum x_i}{n}, \quad s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1}.$$

$$s = \sqrt{s^2}, \quad \text{SE}(\bar{x}) = \frac{s}{\sqrt{n}}.$$

One-sample t-test

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad \nu = n - 1.$$

Use one numerical sample, a fixed reference mean, and unknown population SD. Exact inference uses an independent normal sample.

Two-sided confidence interval

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}.$$

This interval corresponds to a two-sided level- α test of a null mean.

Error probabilities

α = controlled Type I error level, β = Type II error probability at a specified alternative.

$$\text{Power} = 1 - \beta.$$

Remember: s is an SD; s^2 is a variance; $s/\sqrt{n}$ is an SE. The reference value μ_0 is supplied by the null hypothesis, not by replacing it with the observed mean.

19.2 Paired and Two-Sample Tests

Paired t-test

Choose and state the subtraction order. For n complete pairs,

$$\bar{d} = \frac{\sum d_i}{n}, \quad s_d^2 = \frac{\sum d_i^2 - (\sum d_i)^2/n}{n-1}.$$

$$t = \frac{\bar{d} - \Delta_0}{s_d/\sqrt{n}}, \quad \nu = n - 1.$$

For no mean change, $\Delta_0 = 0$. Normality concerns the differences.

Pooled independent-sample t-test

Assume independent samples and equal population variances.

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{s_p \sqrt{1/n_1 + 1/n_2}}, \quad \nu = n_1 + n_2 - 2.$$

Welch independent-sample t-test

Equal population variances are not assumed. Write $a = s_1^2/n_1$ and $b = s_2^2/n_2$.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{\sqrt{a + b}}, \quad \nu \approx \frac{(a + b)^2}{a^2/(n_1 - 1) + b^2/(n_2 - 1)}.$$

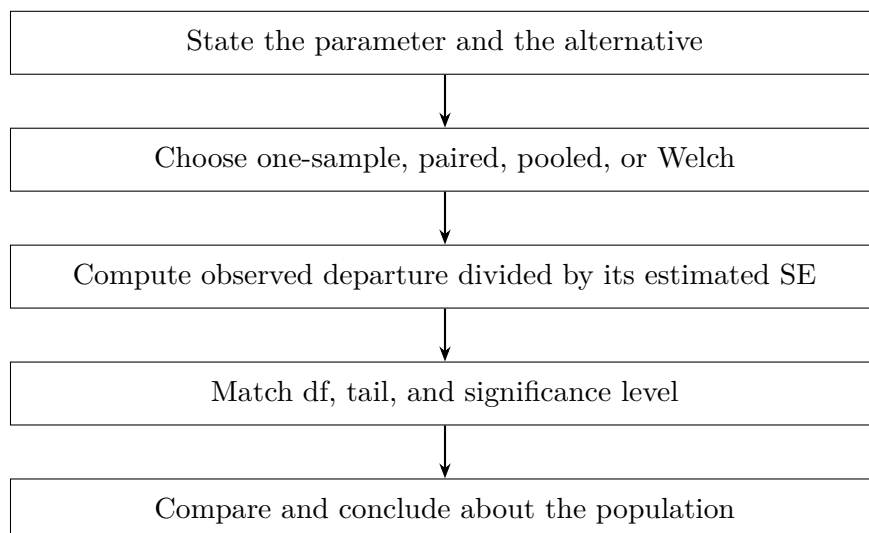
For either independent-sample version, equal-mean testing uses $\Delta_0 = 0$. Always match the standard error and df from the same version.

19.3 Tail Rules and a Compact Decision Map

Throughout these notes, $P(T_\nu > t_{q,\nu}) = q$. The critical values in this notation are positive.

Alternative direction	Rejection rule	p-value
Greater than	$t > t_{\alpha,\nu}$	$P(T_\nu \geq t)$
Less than	$t < -t_{\alpha,\nu}$	$P(T_\nu \leq t)$
Different from	$ t > t_{\alpha/2,\nu}$	$2P(T_\nu \geq t)$

The table uses strict critical inequalities, as is conventional for a continuous distribution. Including the exact boundary leaves the theoretical error probability unchanged. The p-value rule is commonly stated as $p \leq \alpha$; never resolve an apparent boundary using rounded numbers alone.



Useful memory aid: the numerator says *what changed*; the denominator says *how uncertain that change is*; the tail says *which change the question is asking about*.

20 Final Problem-Solving Checklist

Before submitting a solution, ask:

1. Have I identified the actual population and numerical variable?
2. Have I distinguished a fixed benchmark from a second sample mean?
3. Is the design one-sample, paired, or independent-sample?
4. Are my hypotheses about a population parameter?
5. Does H_1 express the question asked, with the correct direction?
6. Have I included equality in H_0 and used its correct boundary?
7. Have I stated the significance level and the relevant assumptions?
8. Is the denominator a standard error, with the correct variance estimate?
9. For pairs, did I define the subtraction order and use the number of pairs?
10. For independent samples, have I justified the pooled or Welch choice?
11. Do the df and critical-value column match the chosen test and tail?
12. Does the sign of t agree with the signed numerator?
13. Have I kept enough precision for the comparison?
14. Does the conclusion state evidence or insufficient evidence in context?

Final reminder
Question $\longrightarrow$ Hypotheses $\longrightarrow$ Correct t-test $\longrightarrow$ Evidence-based conclusion. The purpose of the calculation is to answer the original question about the population mean or mean difference. Every formula should serve that purpose.

Reference checks. The technical formulas and critical-value conventions were checked using the NIST/SEMATECH e-Handbook: two-sample t formulas and Student's t critical values. The explanations, teaching examples, diagrams, and practice questions here are written for these notes. All numerical datasets are teaching examples.